Design of high Frequency Digitally Controlled Oscillators using Machine Learning

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Aim of Work

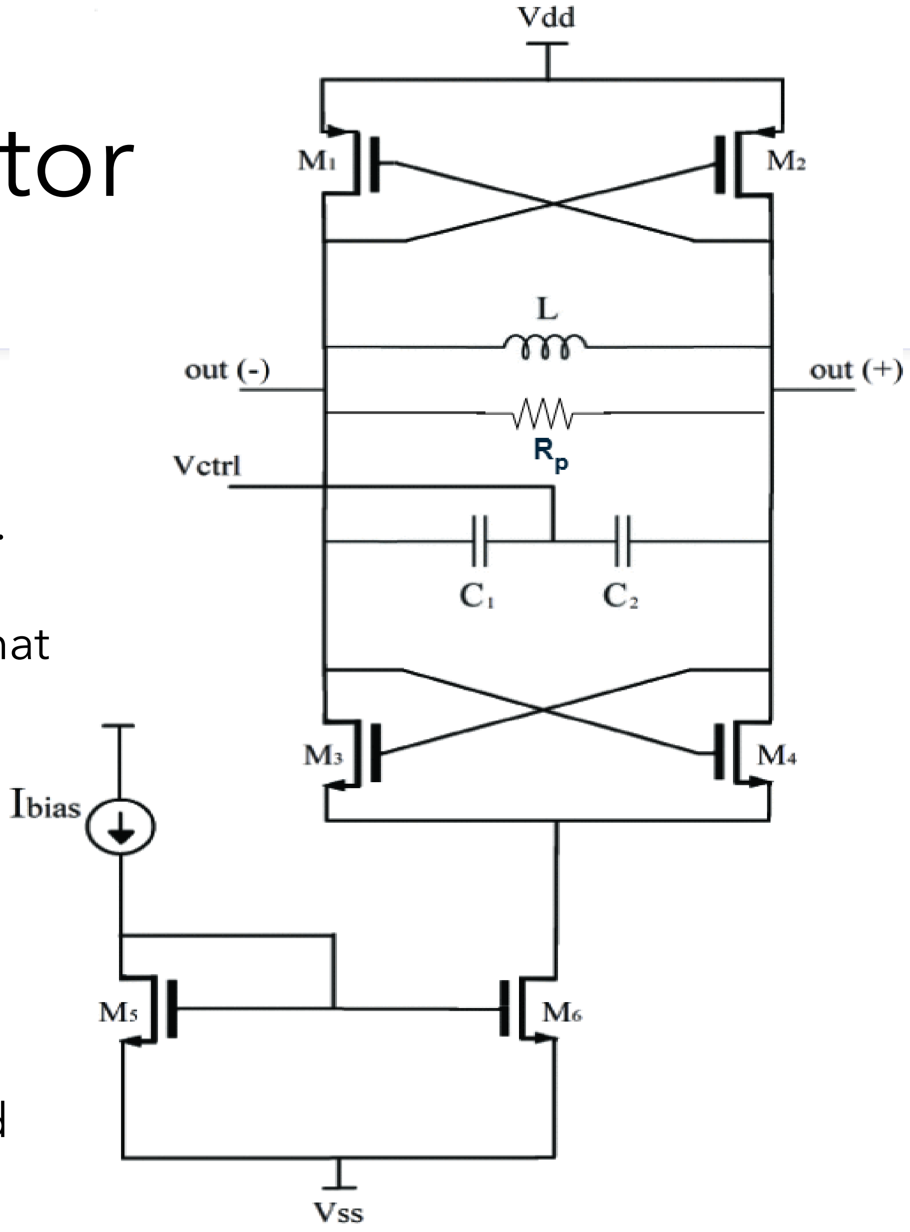
- ✓ Design of high-frequency DCO targeting minimized phase noise and maximized tuning range.
- ✓ Development of a Machine Learning framework that automates the DCO design process with minimized phase noise.
- ✓ Physical implementation and layout optimization, incorporating EM extraction through RaptorX.

Voltage Controlled Oscillator

- The core of the circuit is the LC tank that generates damped oscillations due to the parasitic resistance R_p (lossy resonator).
- The cross-coupled MOS pairs serve as a negative resistance that compensates for the energy loss.
- The circuit produces the oscillation signal at a frequency:

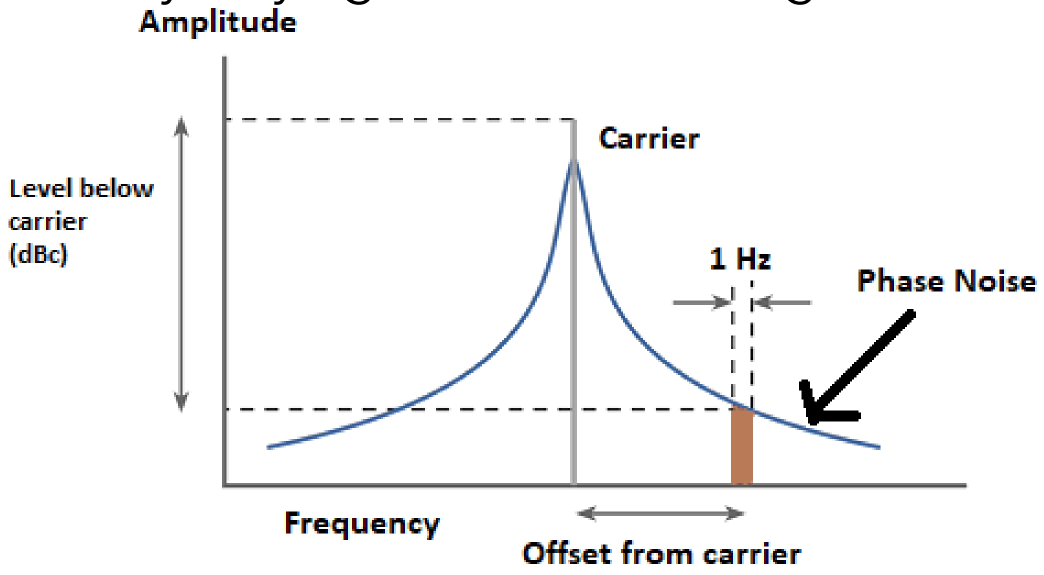
$$f_{osc} = \frac{1}{2\pi\sqrt{L\,(C + C_P)}}$$

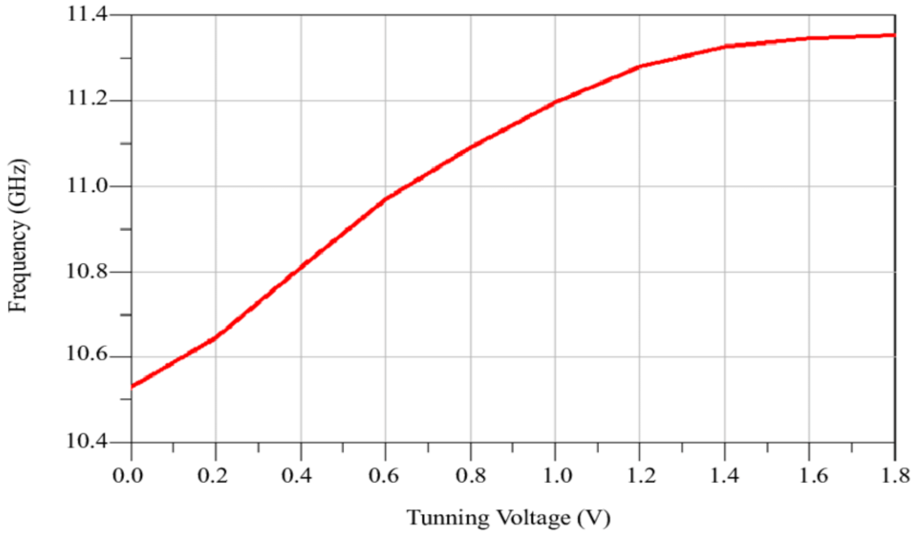
where, C_P is the parasitic capacitance (LC-tank and cross-coupled transistor pairs).



VCO Performance Metrics

- Phase noise is the primary type of noise in oscillators, reflecting frequency variations in the signal. It is frequency-dependent and it is a measure of spectral purity.
- The tuning range of a VCO refers to the frequency span over which the oscillator can be adjusted by varying the control voltage of the varactors.



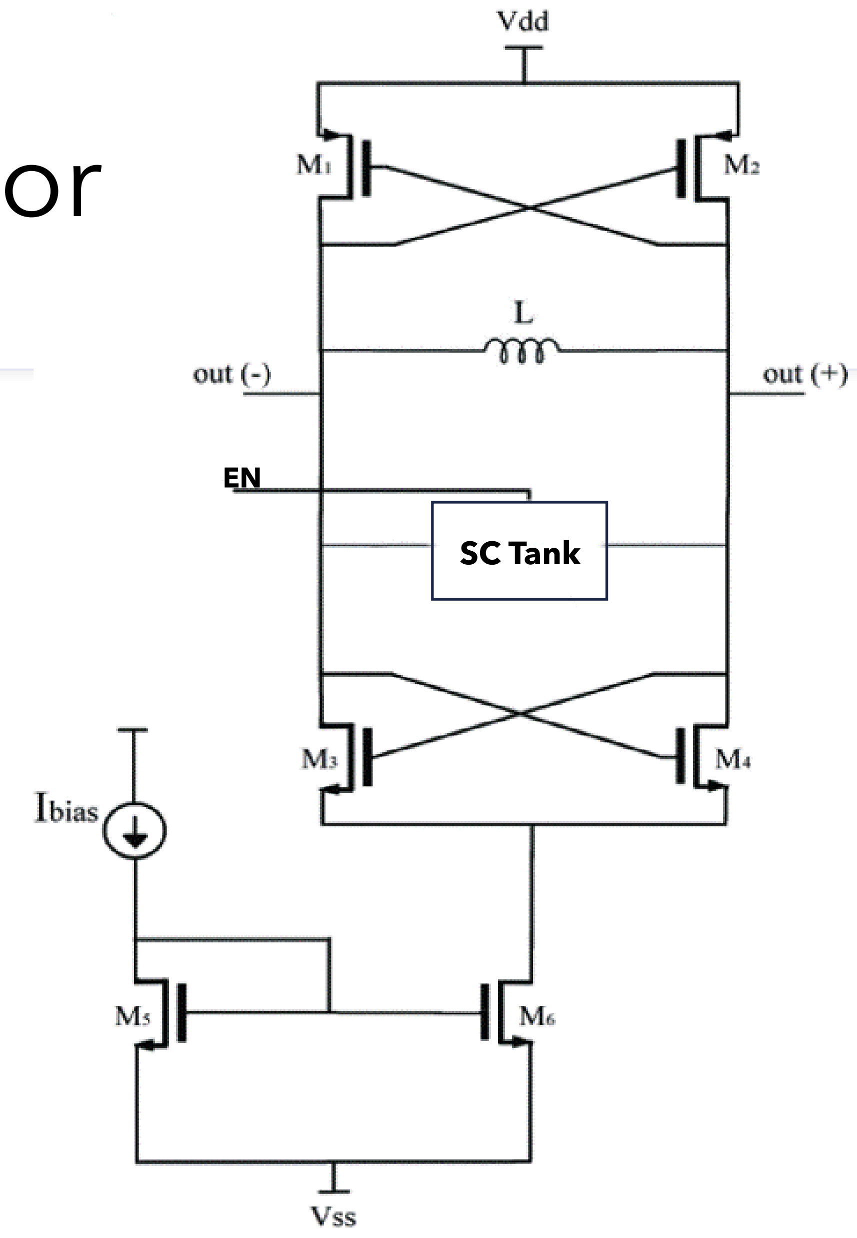


Digitally Controlled Oscillator

- The varactors of the LC VCO are replaced by switchable capacitors

- The transition from VCOs to DCOs is driven by the need for better integration with digital systems and higher tuning range ability.

- The performance of a DCO heavily depends on the choice of the switched capacitor circuit, which eventually determines the tuning range of the oscillator.



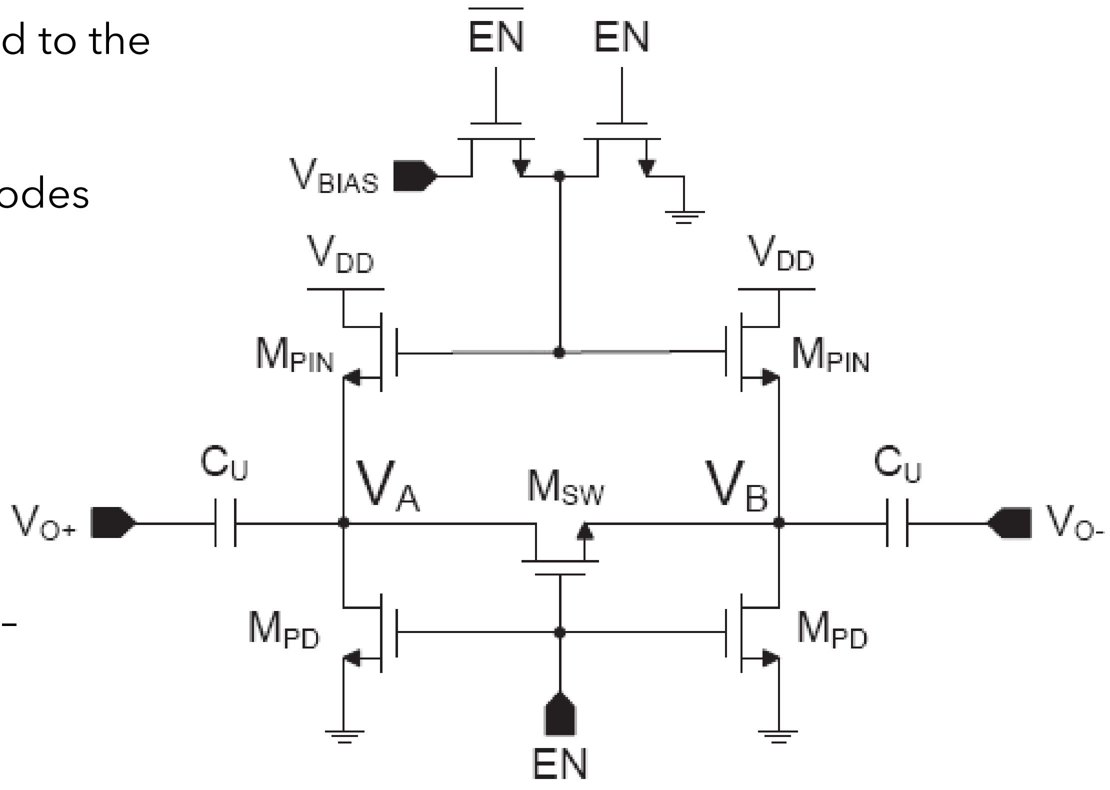
State-of-the-Art Switched Capacitor Circuit

EN = 1:

- Switch M_{SW} differentially shorts V_A and V_B and capacitors C_U add to the overall tank capacitance.
- Transistors M_{PD} are also on, thus pulling down the DC level at nodes $V_{A,B}$, ensuring maximum conductance of M_{SW}

EN = 0:

- Switch M_{SW} is open and nodes V_A and V_B appear as high impedance floating nodes. Capacitors C_U do not add to the overall tank capacitance.
- Transistors M_{PIN} are now conducting, forcing $V_{A,B}$ to equal $V_{BIAS} - V_t$.
- This ensures that MSW will remain off, maintaining a good off-state quality factor



Machine Learning Framework

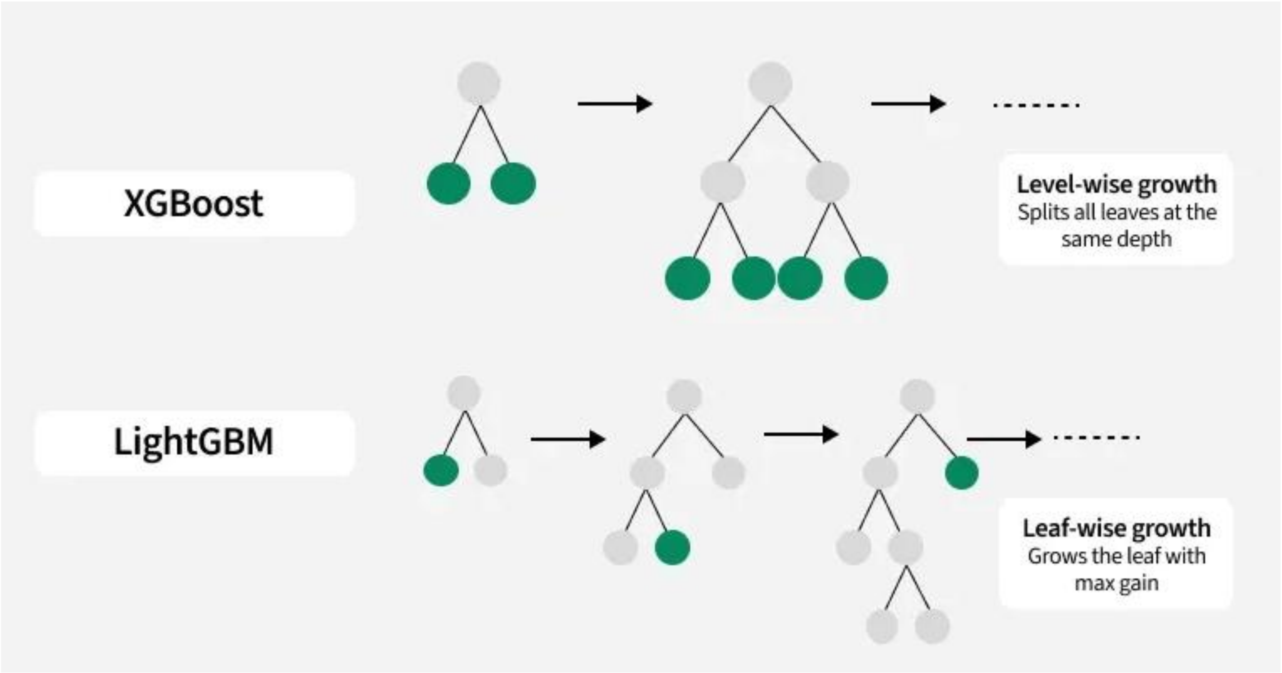
- ✓ Given that VCO and DCO are identical except for the tank, VCO measurements inform the ML framework enabling automated DCO synthesis with minimized phase noise.

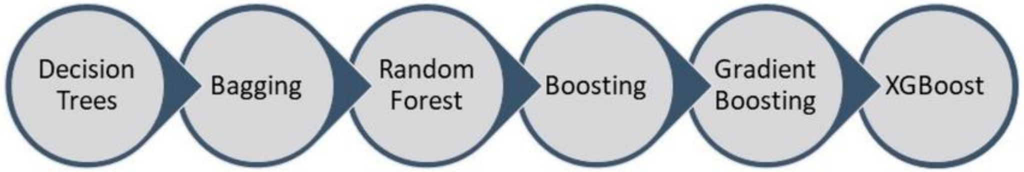
- ✓ The tank-related differences are addressed by characterizing the varactors and switched capacitors as equivalent capacitances through Y-parameter measurements.

- ✓ A dataset was constructed through systematic parameter sweeps resulting at approximately 450 schematic-level VCO designs by varying key circuit parameters. Three 6-bit LC DCO were designed to validate the ML framework

XGBoost and LightGBM

- XGBoost: A regularized gradient-boosting framework over decision trees using second-order (Newton) updates and level-wise growth; emphasizes stability and fine-grained overfitting control, making it strong on tabular data at small–to–medium scale.
- LightGBM: A histogram-based gradient-boosting framework with leaf-wise growth and sampling/feature-bundling accelerators. Delivers fast, memory-efficient training that scales to large, high-dimensional datasets while maintaining competitive accuracy.

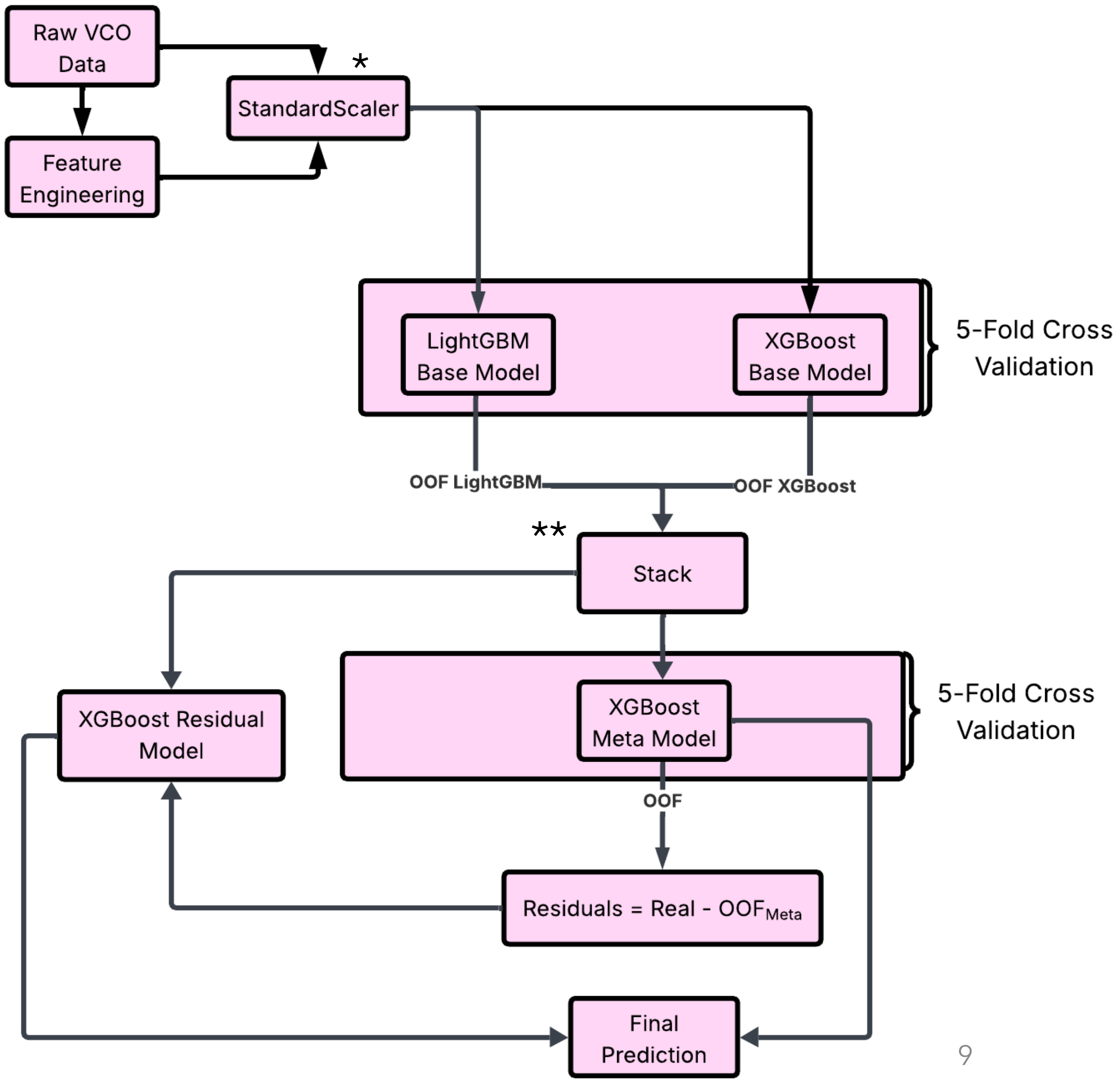




Frequency Prediction Model

Input Parameters

Parameter	Range
NMOS transistor width W_1	3.4 – 50 μm
NMOS transistor length L_1	32 – 400 nm
PMOS transistor width W_2	3.4 – 50 μm
PMOS transistor length L_2	32 – 400 nm
Inductor value L	150 – 1520 pH
Capacitor value C	50 – 2500 fF



*Per-fold preprocessing and feature engineering: A StandardScaler is fit only on the training portion of each fold and then applied to that fold's validation subsets using the same μ, σ.

**Meta-model is trained exclusively on OOF predictions from base models; validation/test predictions are not fed back into training.

Hyperparameter Tuning

For the hyperparameter tuning of all models (base, meta and residual), Optuna optimization framework was employed with 3-fold cross validation.

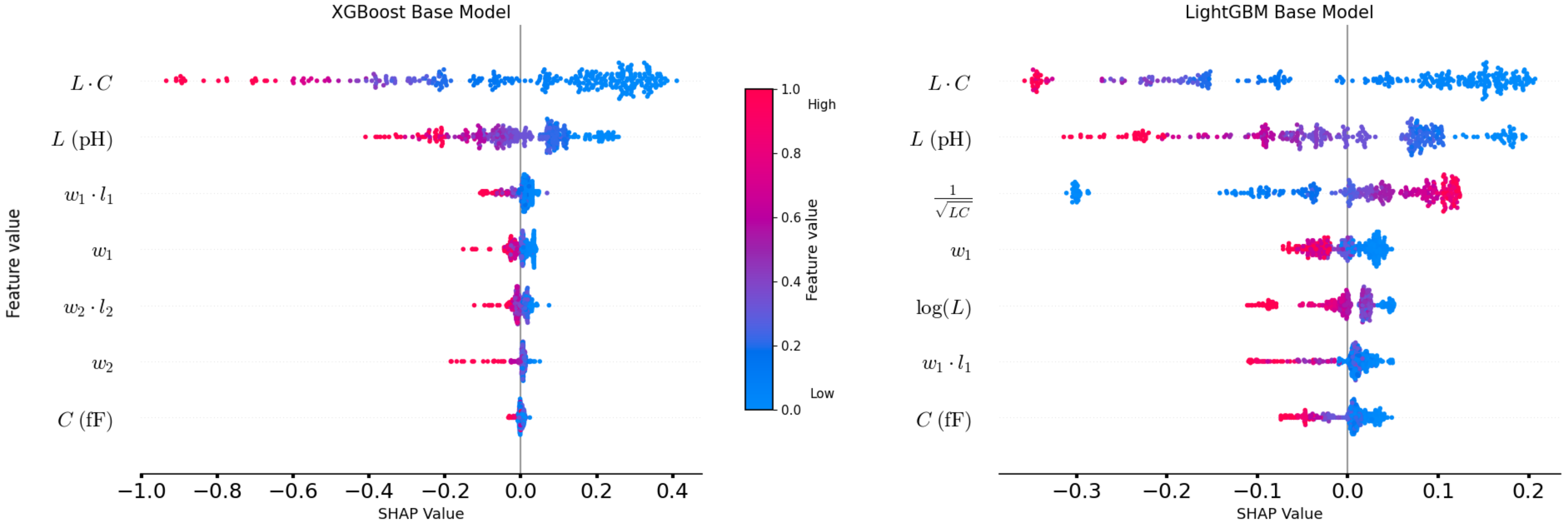
Meta Model Hyperparameters Value

Hyperparameter	Function	Value
learning_rate (eta)	Shrinkage factor controlling the contribution of each new tree	0.12
n_estimators (num_round)	Number of boosting rounds (trees) to train	790
max_depth	Maximum depth of each tree, controls model complexity	12
subsample	Fraction of training samples used for growing each tree	0.65
colsample_bytree	Fraction of features (columns) used when constructing each tree	0.68
gamma (min_split_loss)	Minimum loss reduction required to make a split	0.06

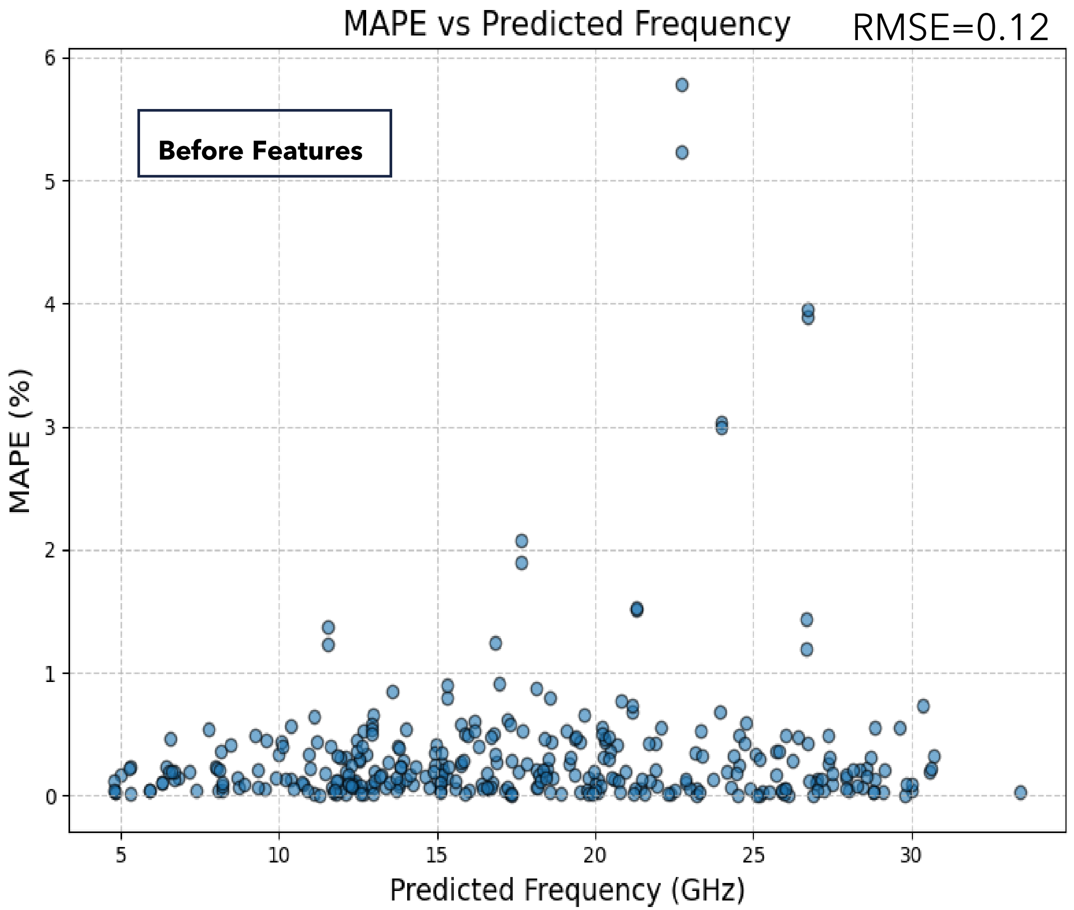
Avoiding Data Leakage

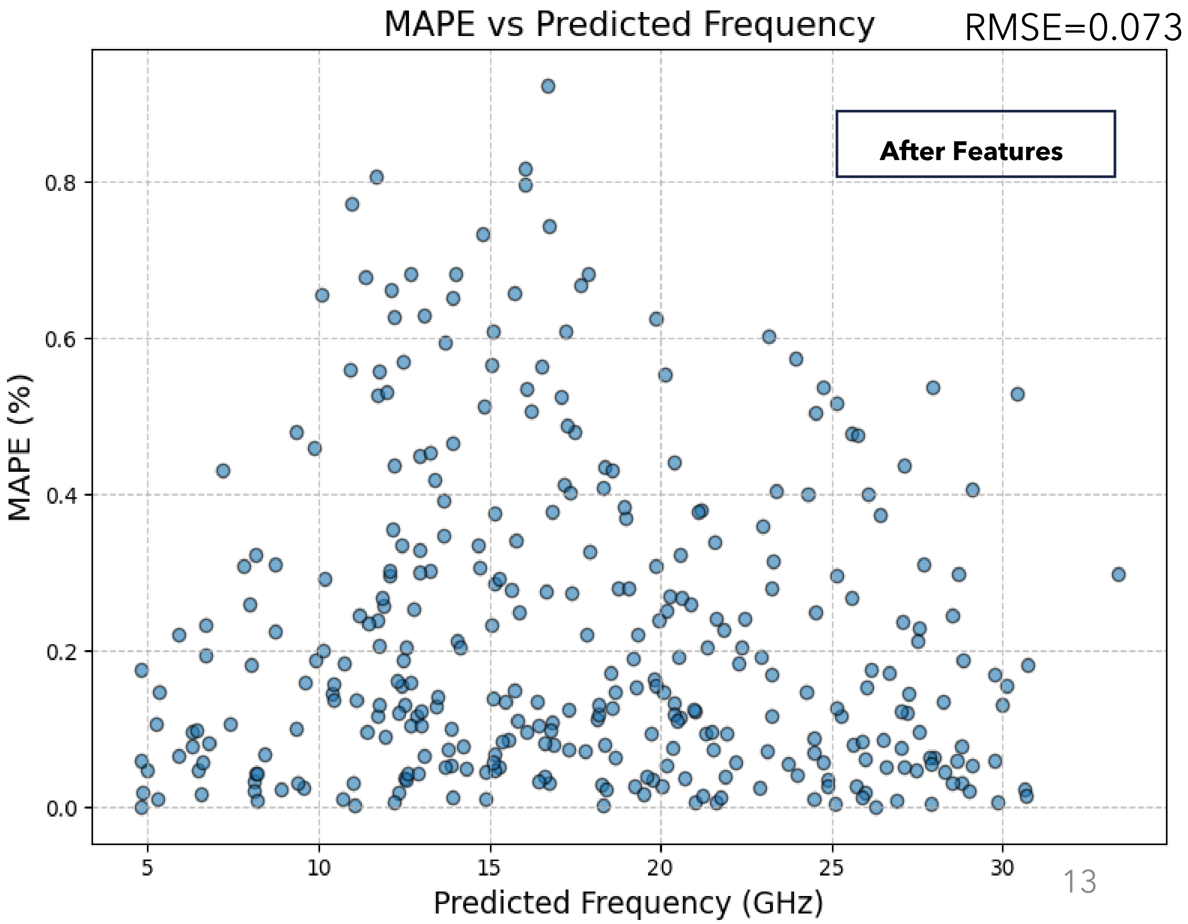
- Per-fold preprocessing and feature engineering: A StandardScaler is fit only on the training portion of each fold and then applied to that fold's validation subsets using the same μ, σ. This ensures no distributional information from validation/test influences training.
- Meta-model is trained exclusively on OOF predictions from base models; validation/test predictions are not fed back into training.
- Nested cross-validation for unbiased model selection is performed. An outer 5-fold CV estimates generalization; an inner 3-fold CV (on the outer-train split) is used by Optuna for hyperparameter optimization. The outer-test split remains untouched during tuning.

Feature Importance

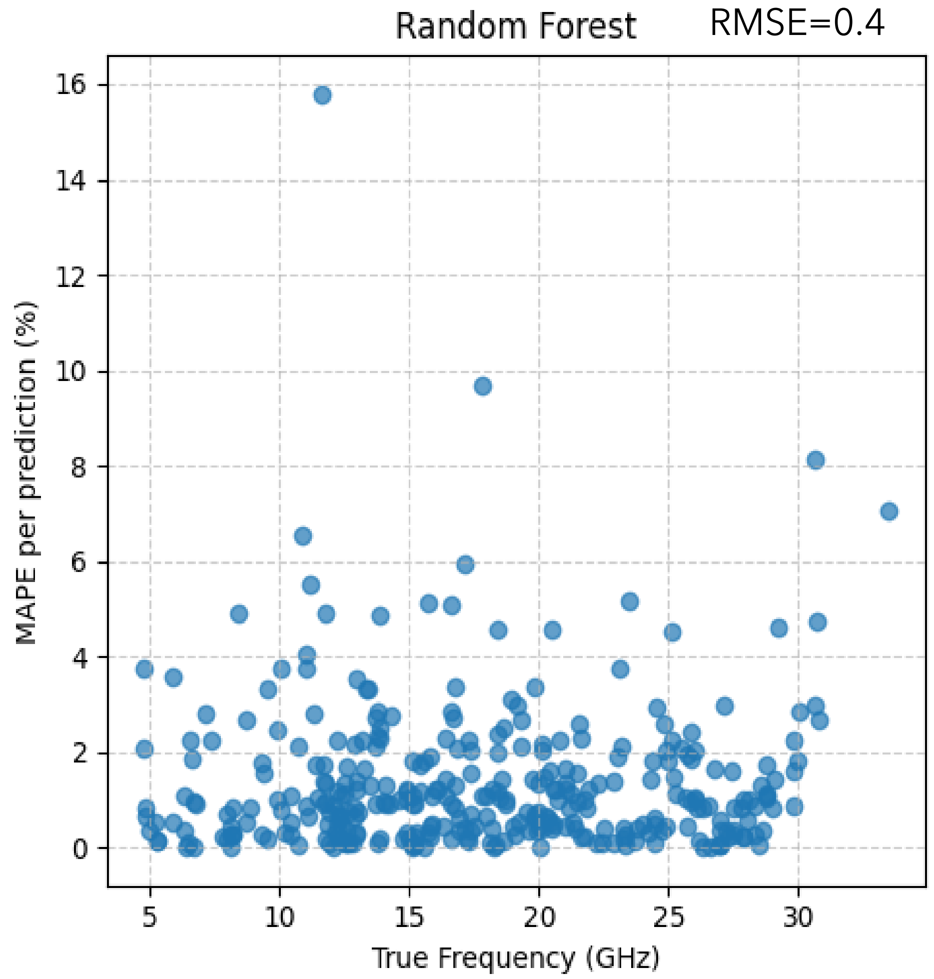


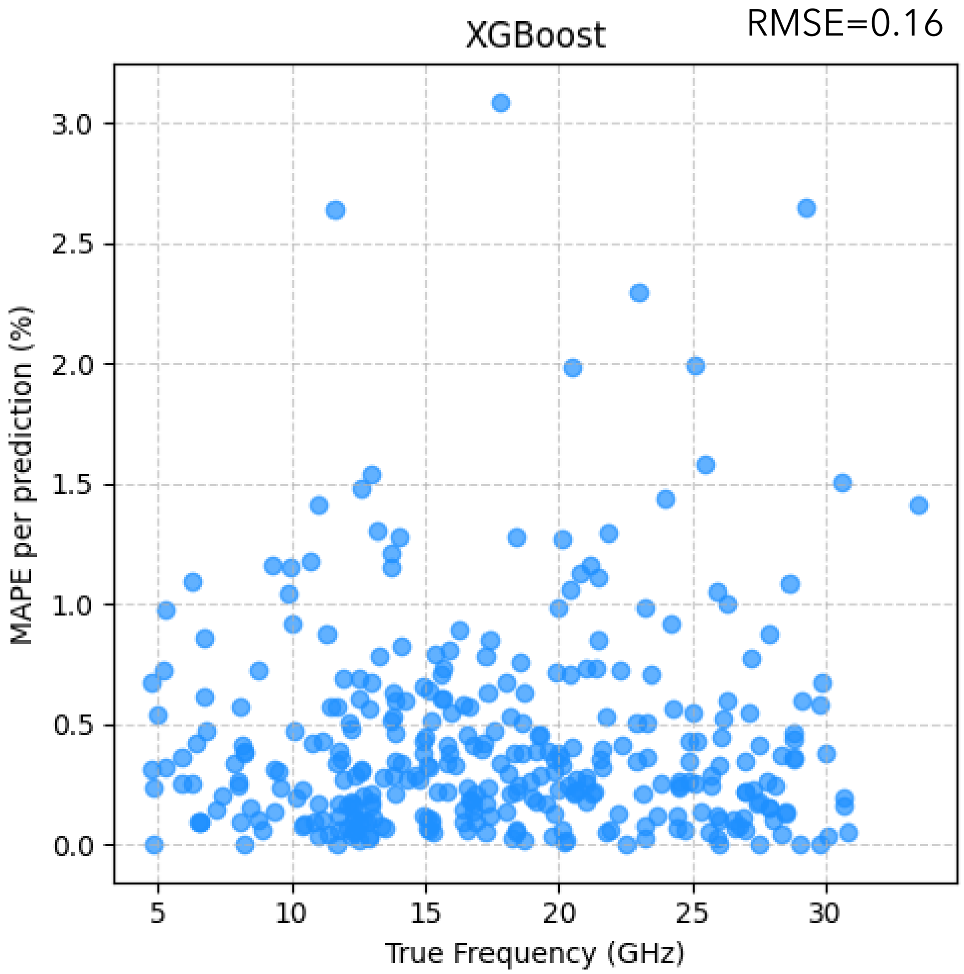
Feature Engineering





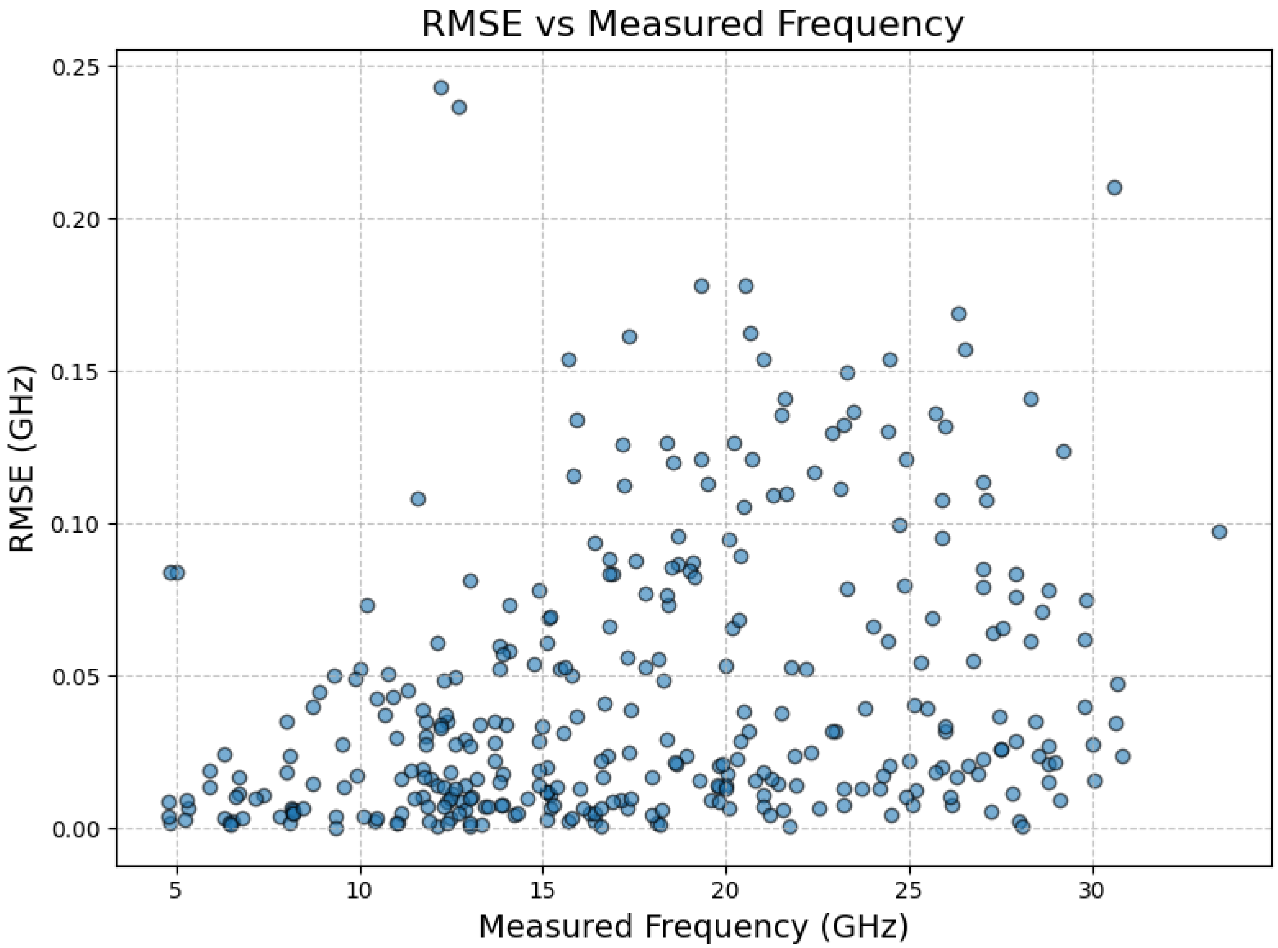
Tree-Based Machine Learning Algorithms Comparison





*For all models, hyperparameter optimization was performed using Optuna combined with 3-fold cross-validation. The seven most influential features were identified and used across all algorithms to ensure consistency in model comparison.

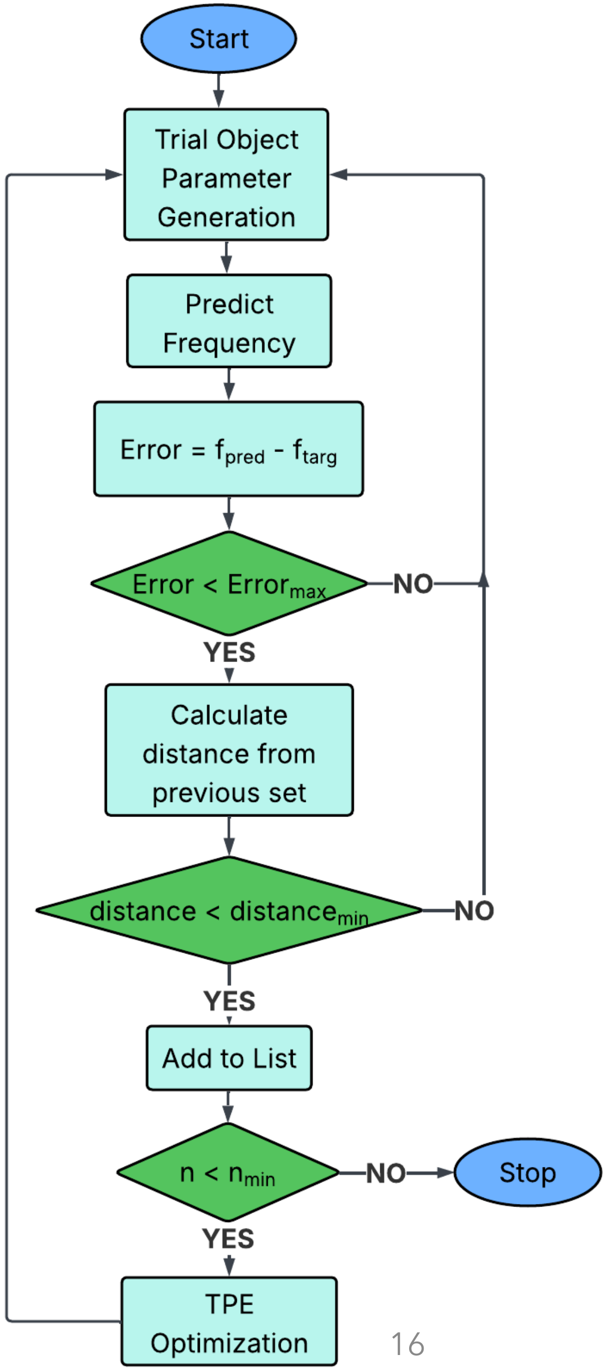
Higher Prediction Error for higher Frequency



- In high frequencies the RMSE per prediction increases compared to lower frequencies.
- The need for band-specific training, therefore, arises.
- However, the data available is not sufficient for separate training.

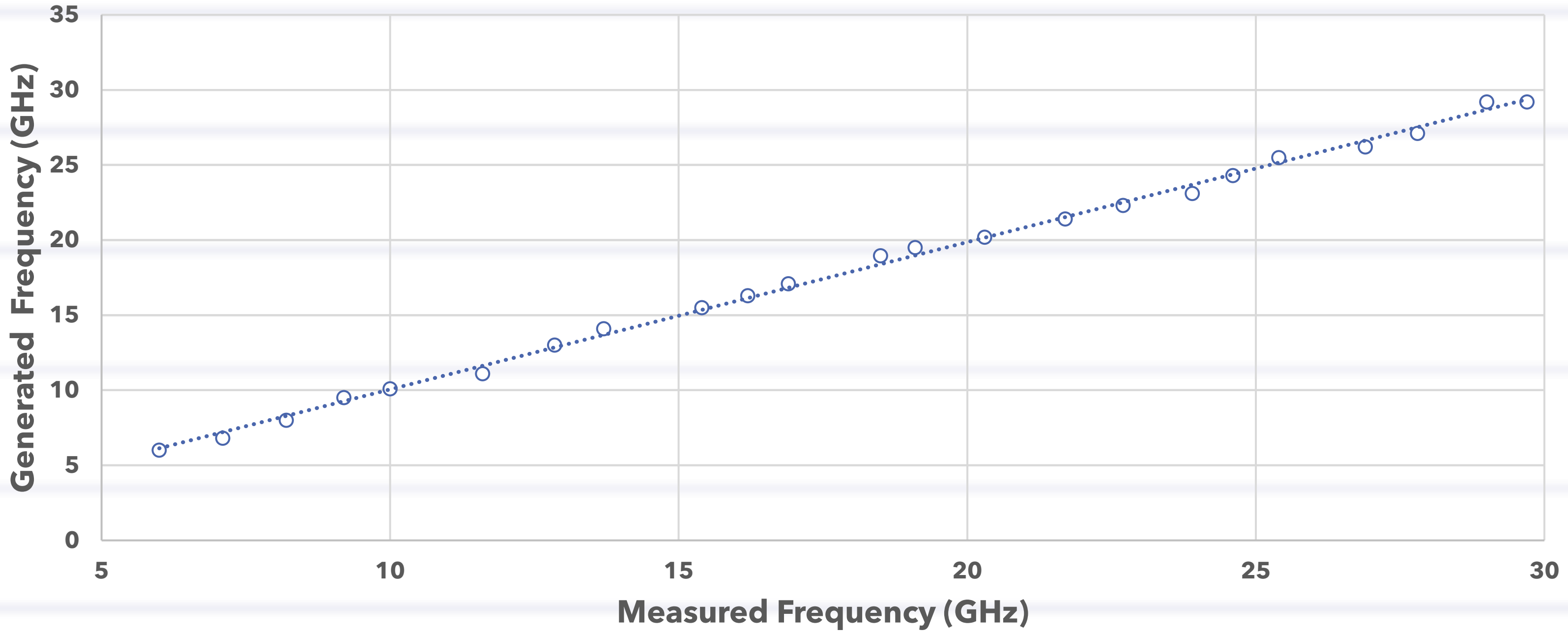
Frequency Generator Algorithm

- Optuna's Trial object is used to suggest potential values for each parameter within predefined ranges.
- The TPE (Tree-structured Parzen Estimator) constructs probabilistic models for the objective function, distinguishing between areas of the parameter space that yield good results (low error) and those that do not.
- The TPE ensures diversity in the search by maintaining an exploration-exploitation balance.



Frequency Generator Validation

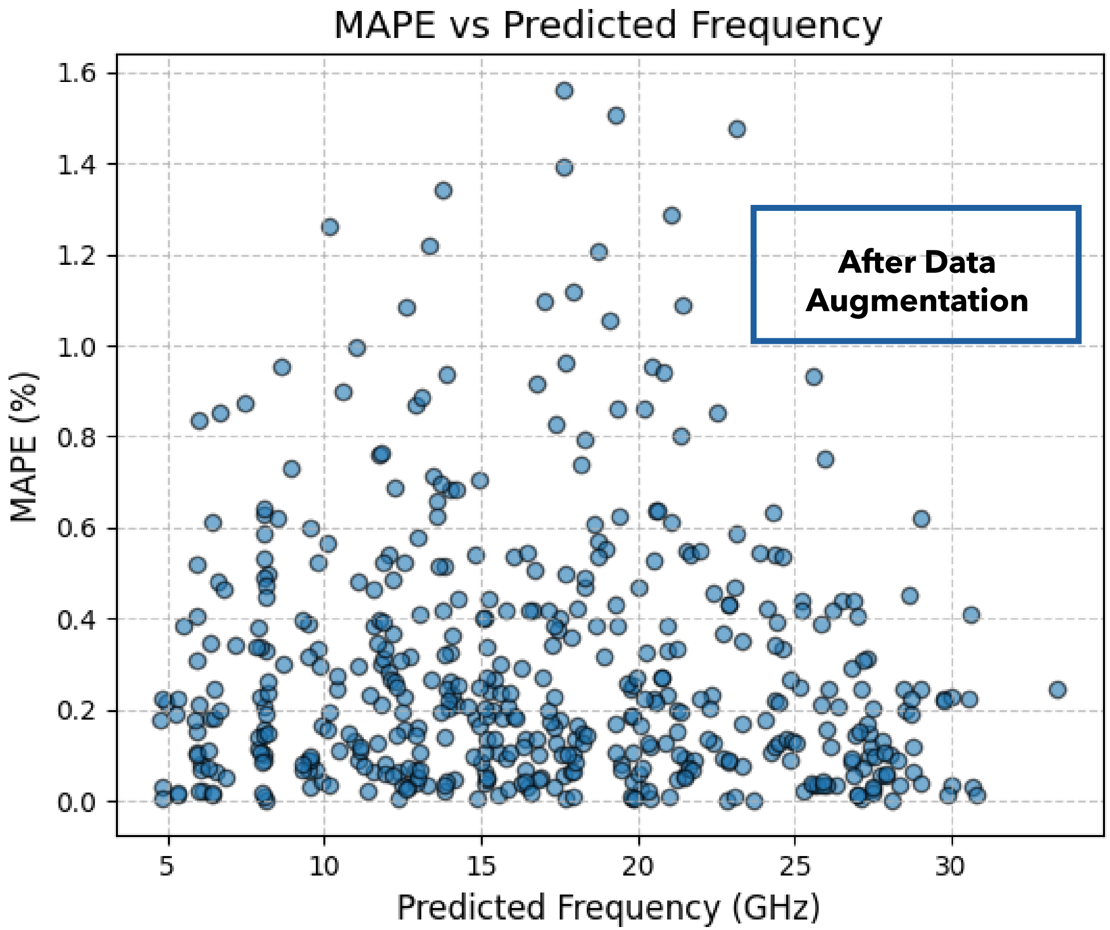
A sample of evenly distributed parameter sets was designed as VCO circuits and tested to validate the accuracy of the frequency generator.



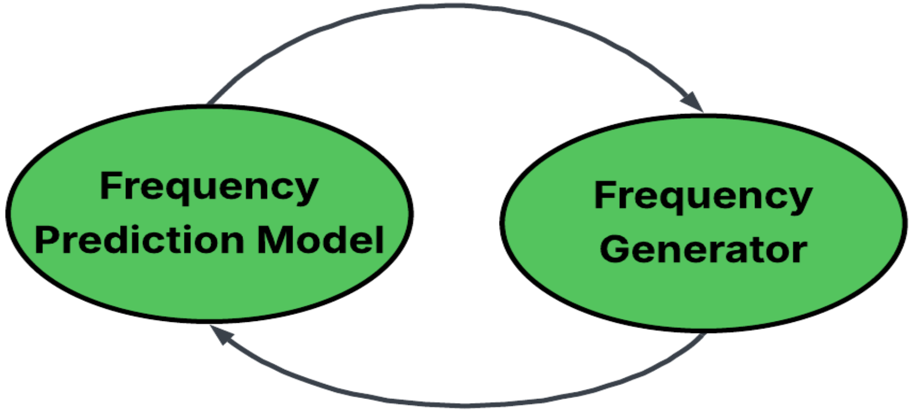
Metric	Value
RMSE*	0.38
MSE*	0.146
MAPE*	1.85%

*For this random sample of generated sets.

Data Augmentation



The final input dataset for the frequency prediction model is composed of 50% simulated data and 50% generated data.



After data augmentation the maximum MAPE per prediction doubled.

Band-Specific Modeling Approach

VCO exhibits different responses across frequency ranges when certain parameters are varied (non-linear). Training in a frequency-aware manner improves the accuracy and robustness of the model :

✓ The input data is divided into three frequency bands: 5-11 GHz (LFB) , 11–20 GHz (MFB) , and 20–30 GHz (HFB).

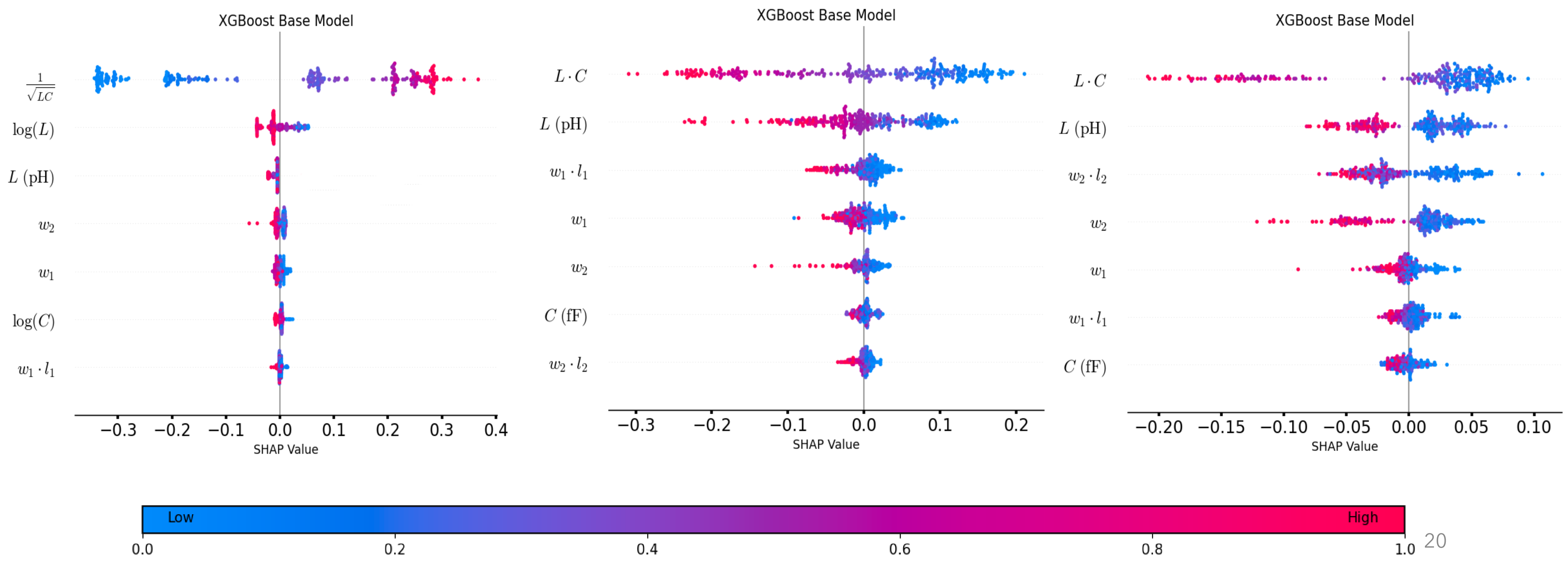
✓ Separate models are trained for each band with tailored feature engineering.

Distinctive Feature Examination

Low Frequency Band

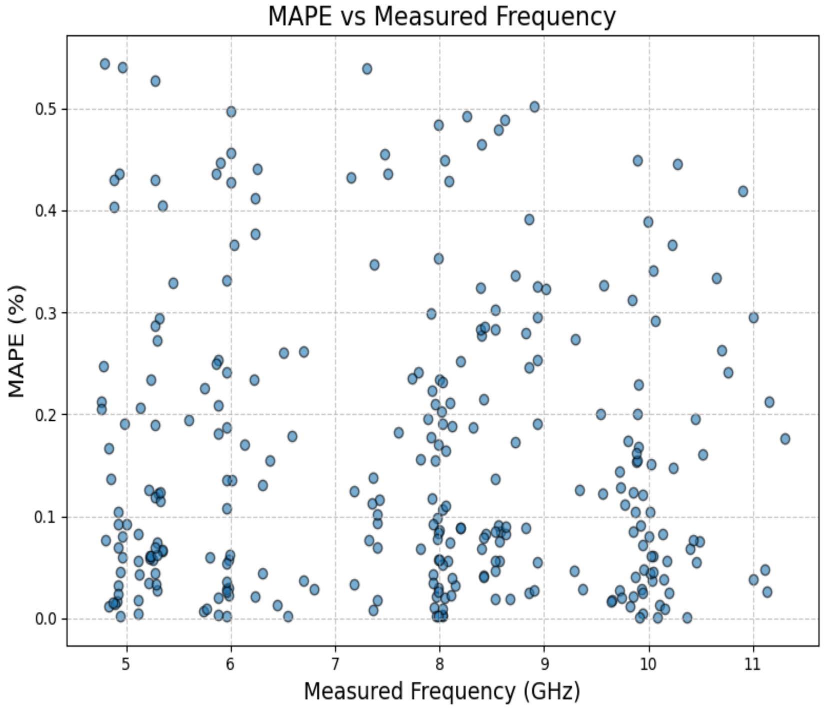
Middle Frequency Band

High Frequency Band



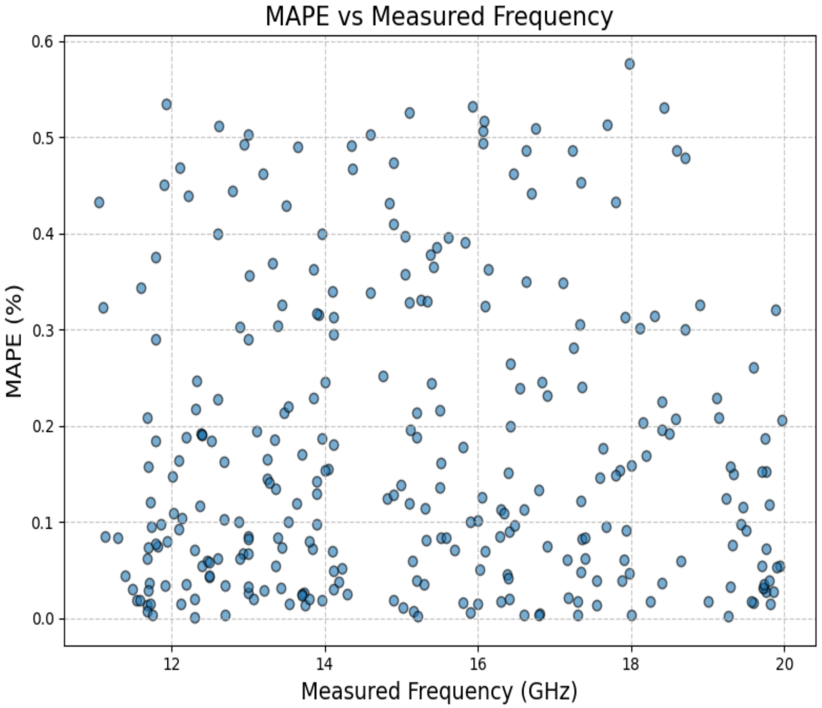
Model Performance

Low Frequency Band



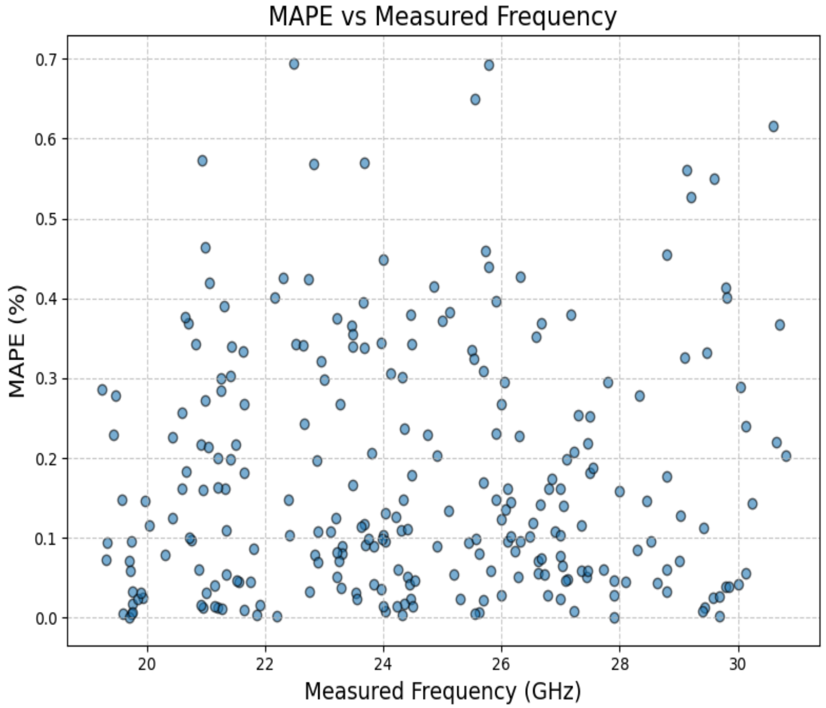
RMSE=0.025 MSE-0.0006 MAPE=0.21. (%)

Middle Frequency Band



RMSE=0.04 MSE-0.002 MAPE=0.2 (%)

High Frequency Band

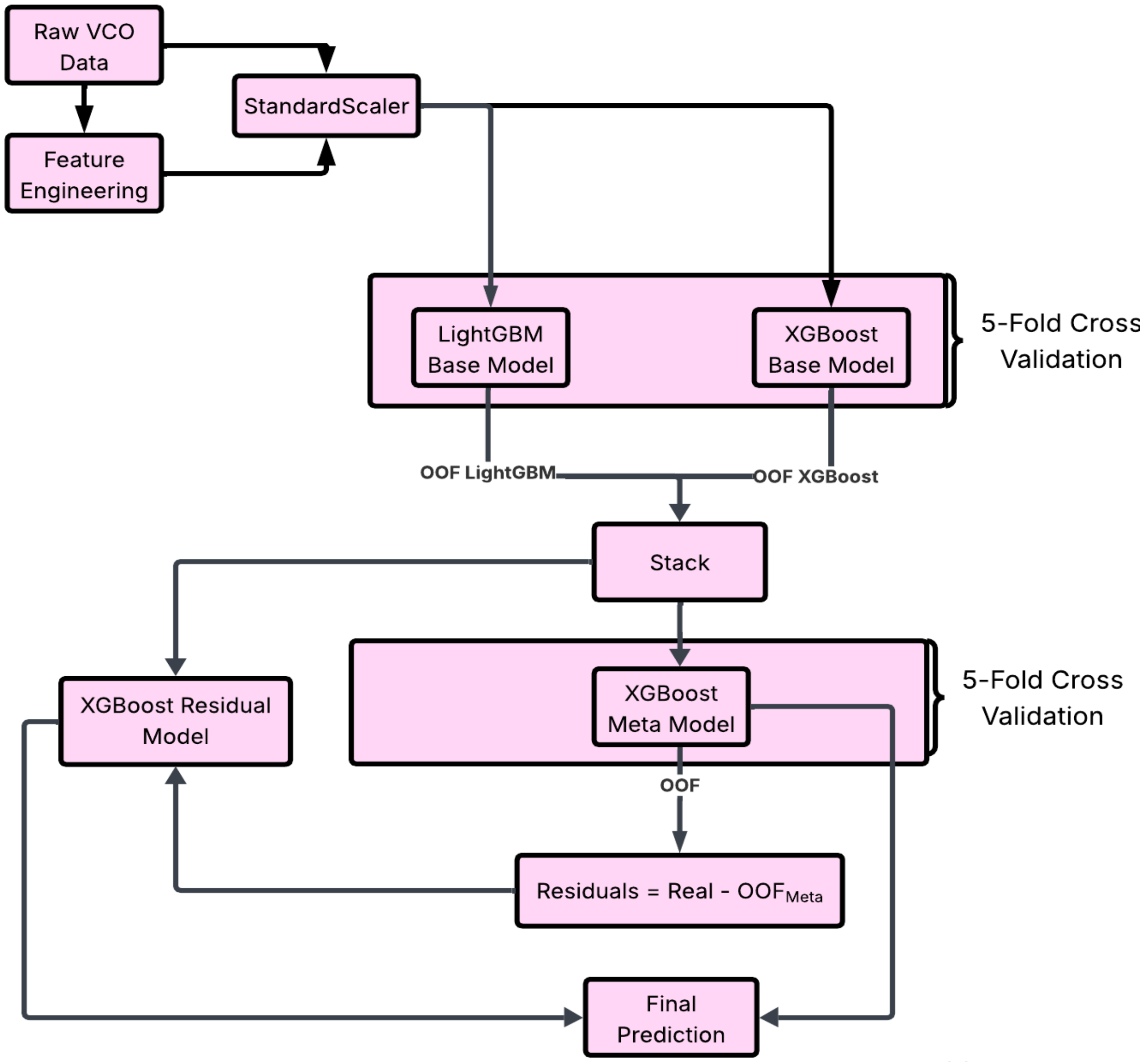


RMSE=0.06 MSE-0.003 MAPE=0.16 (%)

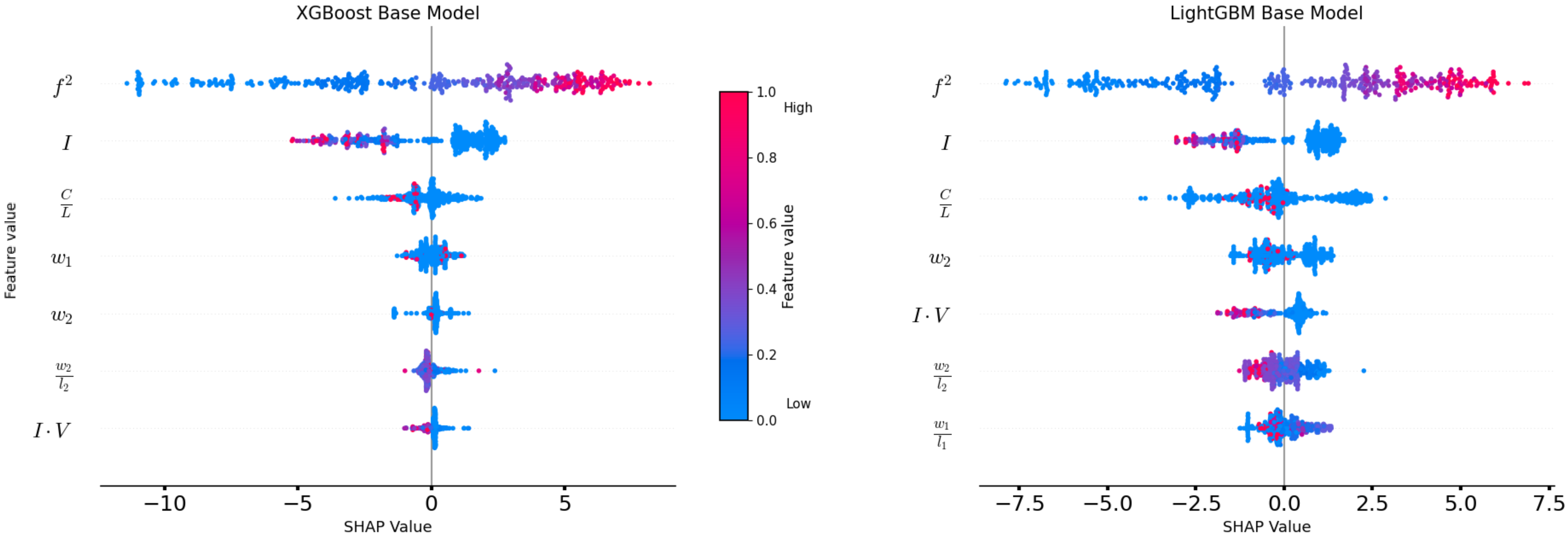
After band specific training the maximum MAPE per prediction is lower than before data augmentation and data separation.

Phase Noise Prediction Model

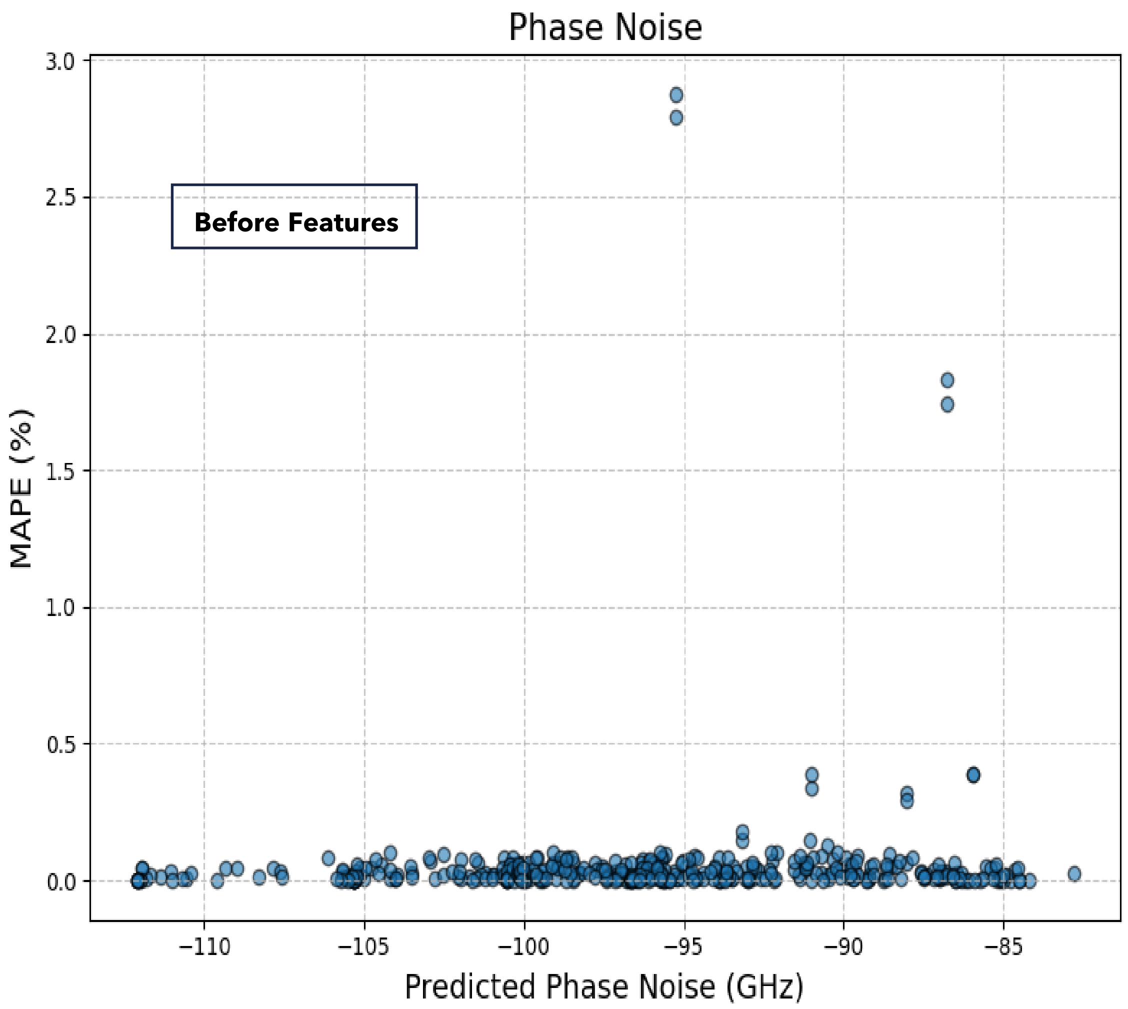
Parameter	Range
NMOS transistor width W_1	3.4 – 50 μm
NMOS transistor length L_1	32 – 400 nm
PMOS transistor width W_2	3.4 – 50 μm
PMOS transistor length L_2	32 – 400 nm
Inductor value L	150 – 1520 pH
Capacitor value C	50 – 2500 fF
Bias current I_{bias}	0.4 – 8 mA
Supply voltage V_{DD}	0.5 – 0.8 V
Frequency F	5 – 30 GHz

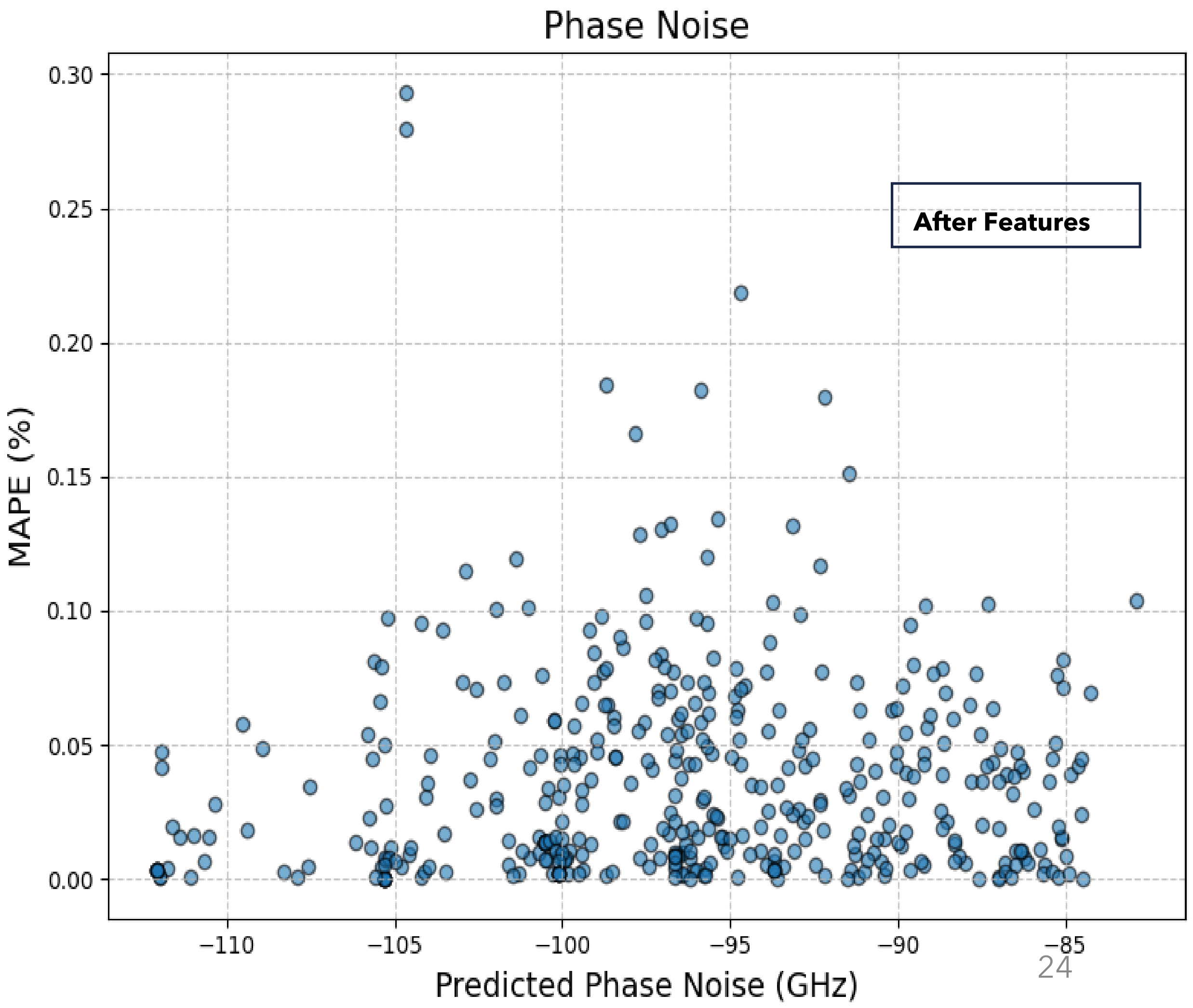


Feature Importance



Feature Engineering

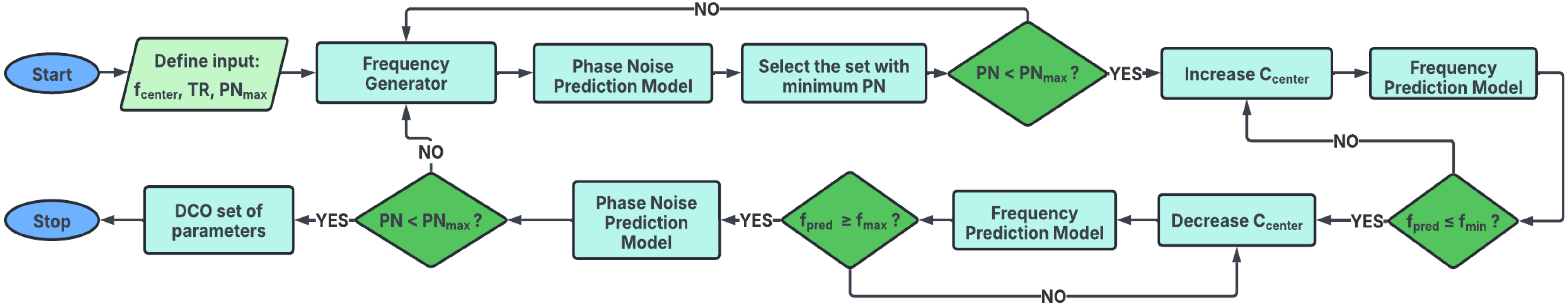




HFB Prediction Model Validation on DCO

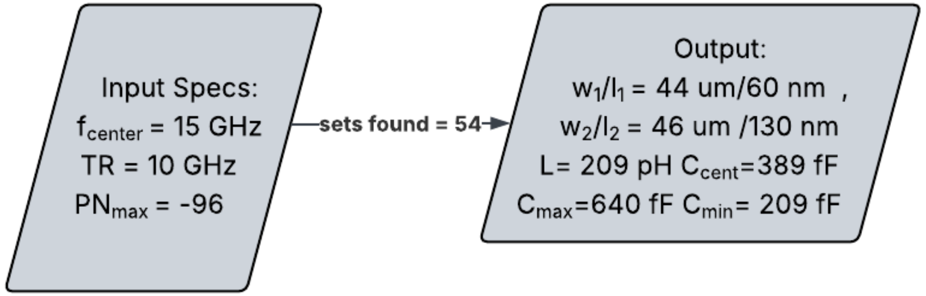
Input Word	Frequency (GHz)	Phase Noise (dBc/Hz)	Predicted Frequency (GHz)	Predicted Phase Noise (dBc/Hz)	Δ (%) Frequency	Δ (%) Phase Noise
000000	30.0	-92.0	29.8	-92.4	0.67	0.43
000001	27.6	-93.5	27.3	-94.1	1.10	0.64
000011	25.4	-95.0	26.0	-96.0	2.31	1.04
000111	23.7	-96.5	24.1	-96.7	1.66	0.21
001111	22.3	-97.7	22.6	-97.8	1.33	0.10
011111	21.0	-98.7	21.0	-99.2	0	0.50
111111	20.0	-99.7	20.4	-99.2	1.96	0.50

Final DCO Generator Algorithm

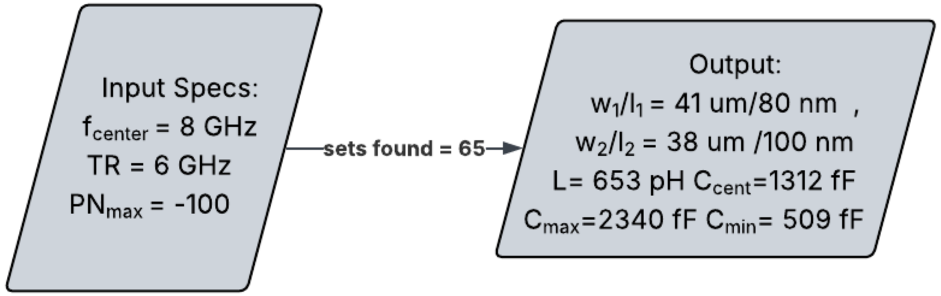


DCO Generator to the Test

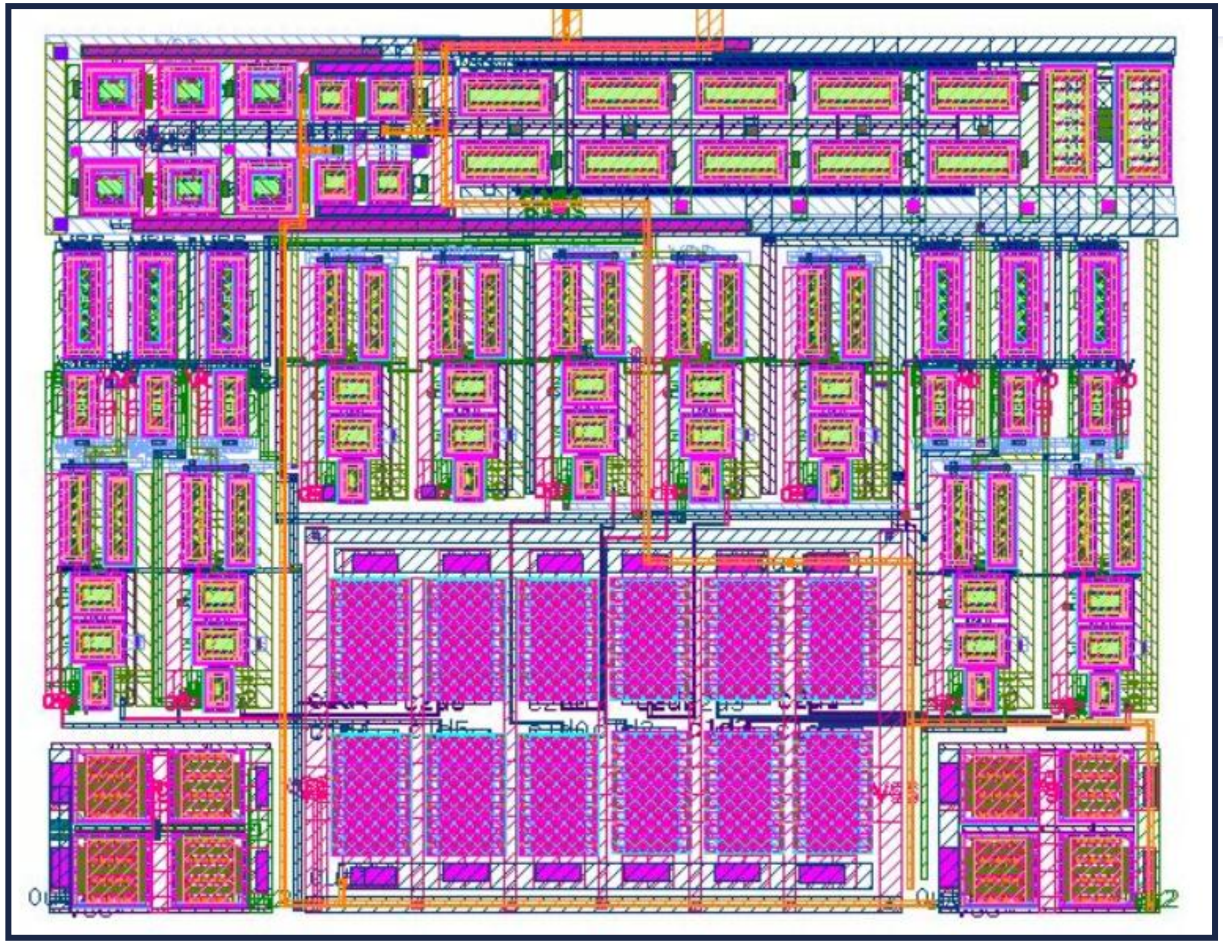
Predicted Frequency (GHz)	Predicted Phase Noise (dBc/Hz)	Measured Frequency (GHz)	Measured Phase Noise (dBc/Hz)	Δ (%) Frequency	Δ (%) Phase Noise
10	-104.6	10.6	-100	6	4.4
15	-100.7	15.1	-98.4	0.67	2.3
20	-97.4	19.5	-94.9	2.5	2.6

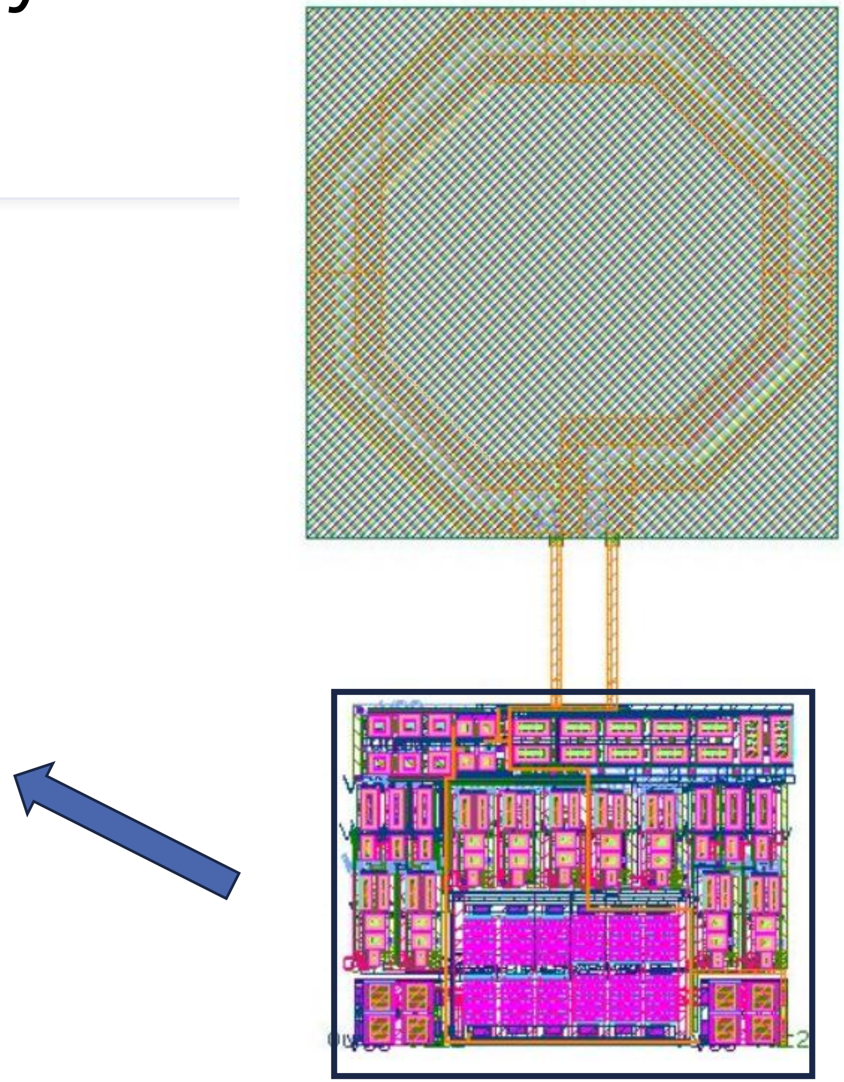


Predicted Frequency (GHz)	Predicted Phase Noise (dBc/Hz)	Measured Frequency (GHz)	Measured Phase Noise (dBc/Hz)	Δ (%) Frequency	Δ (%) Phase Noise
5	-105.6	5.3	-110.0	6.00	4.15
8	-100.7	7.8	-100.2	2.50	0.50
11	-97.4	11.2	-99.0	1.82	1.64

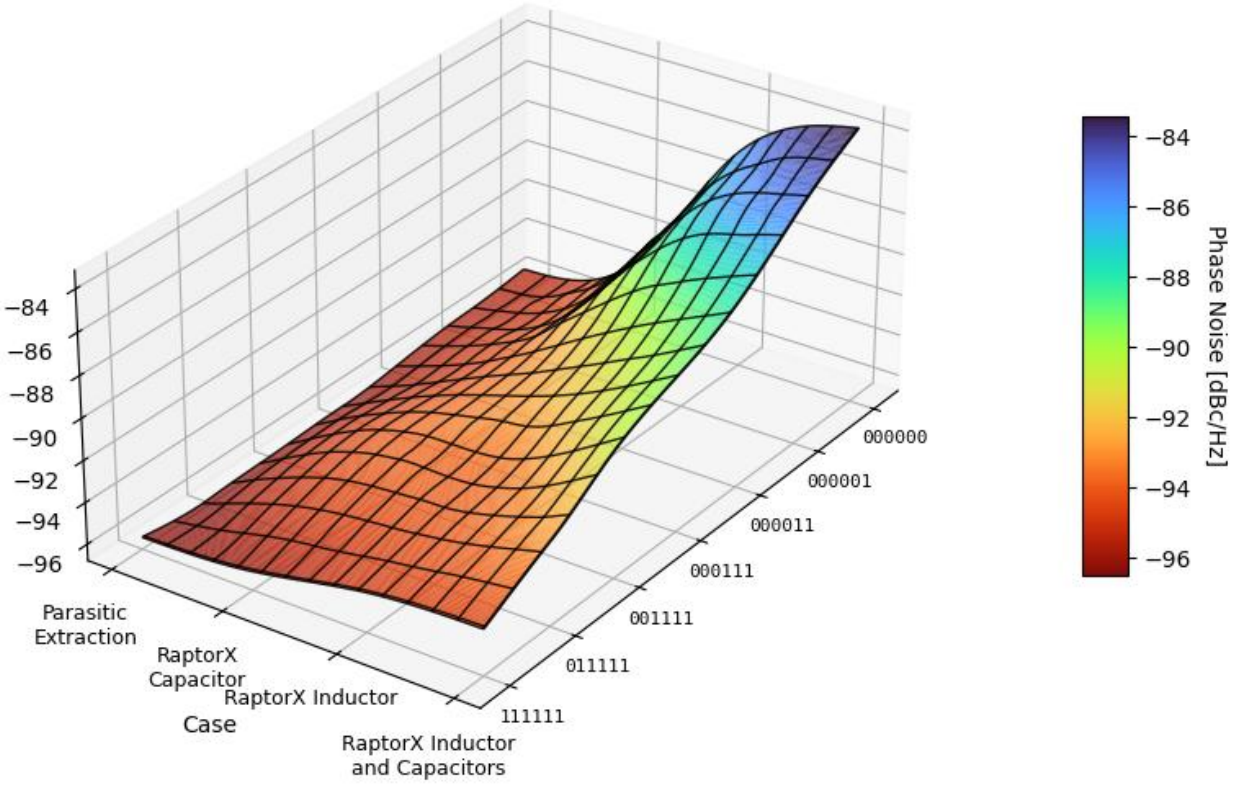


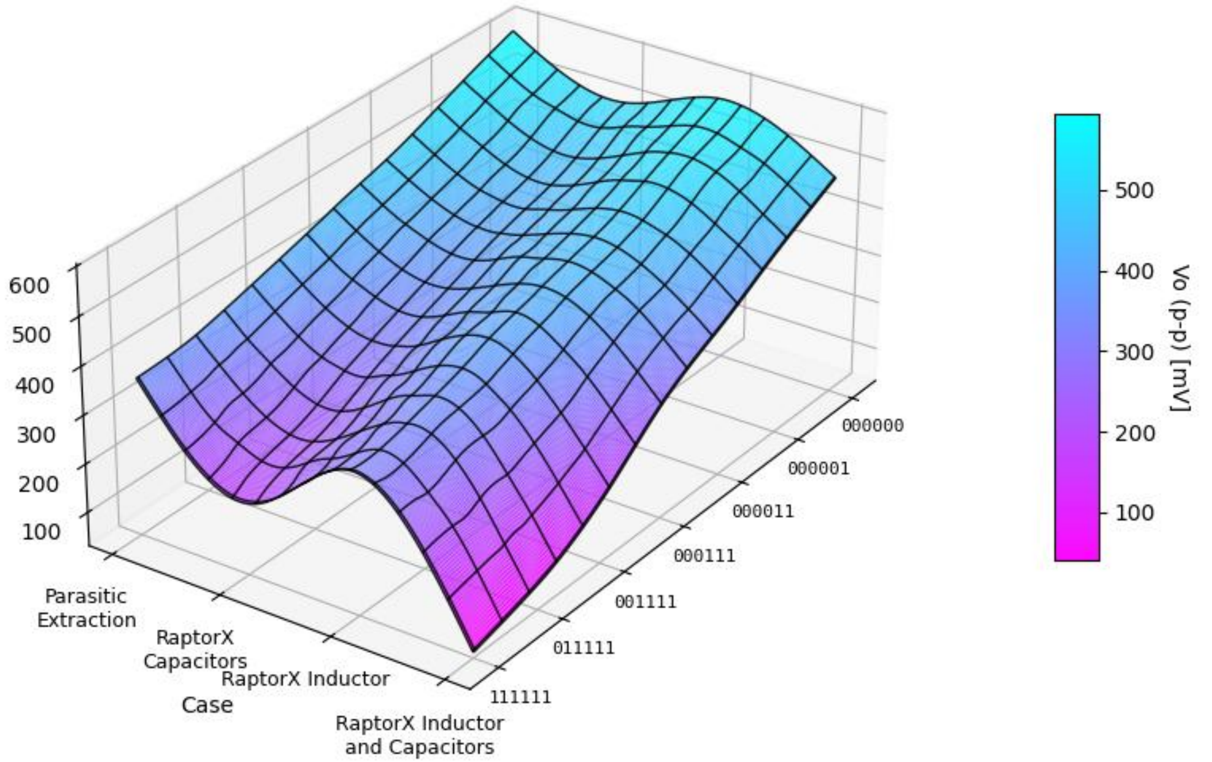
High-Frequency-Band DCO Physical Design



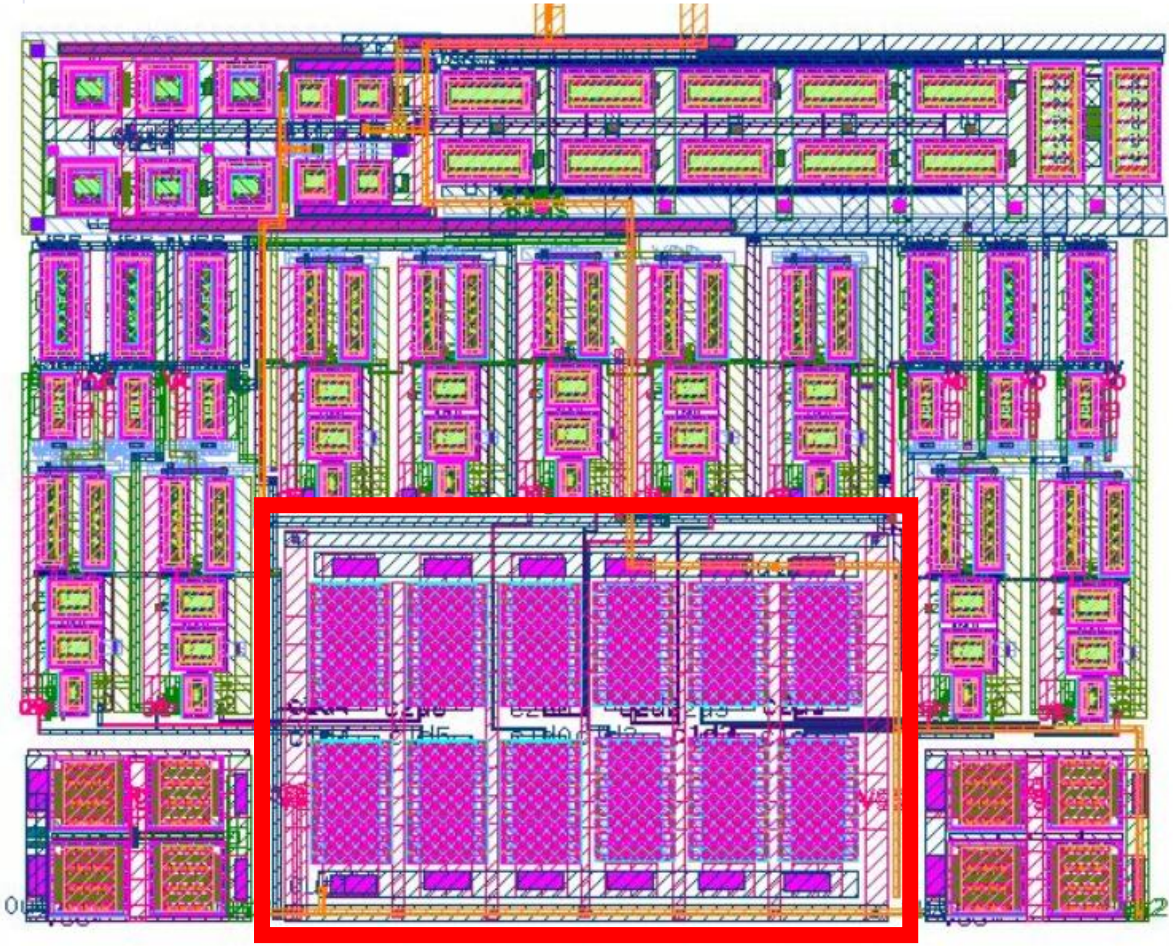


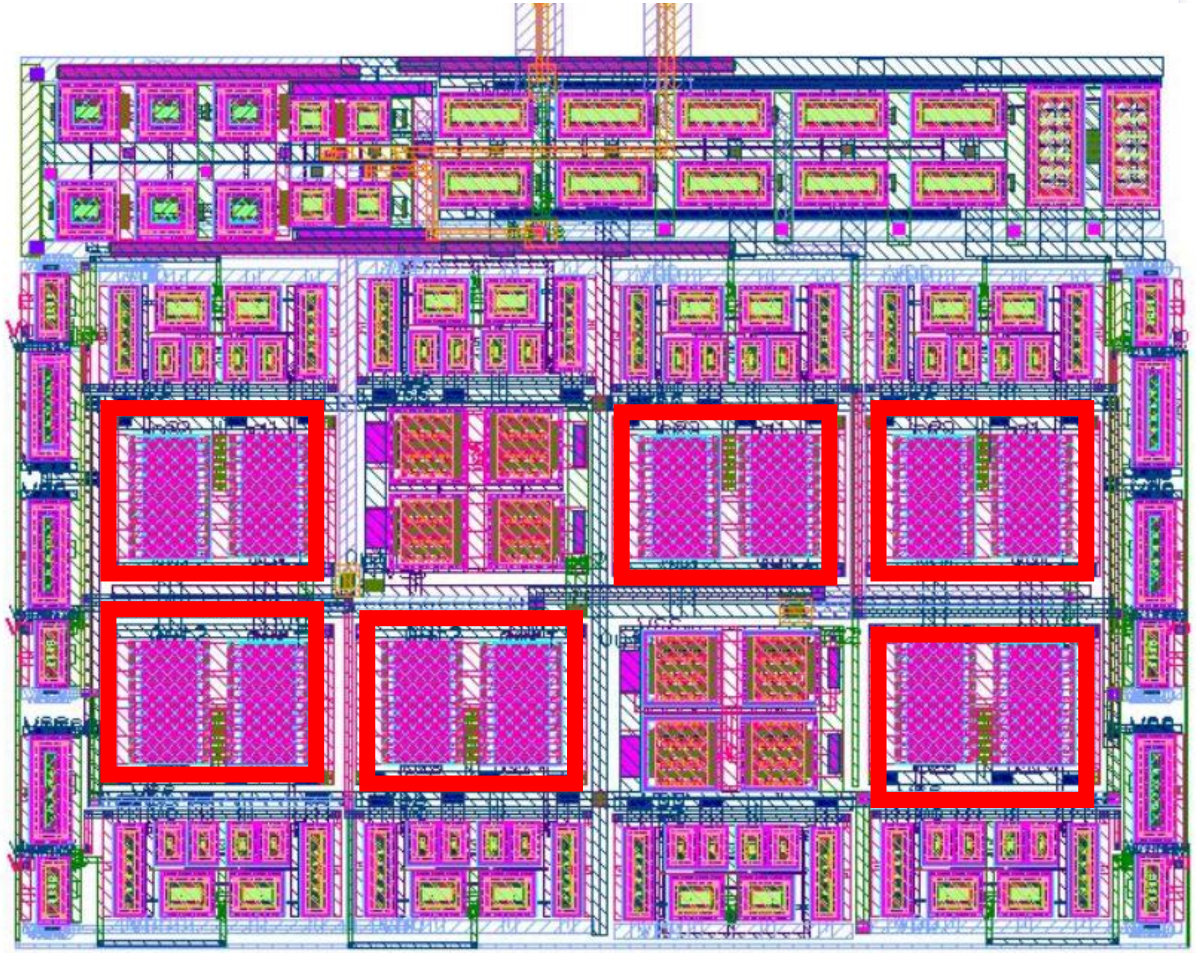
HFB DCO Post-Layout Performance



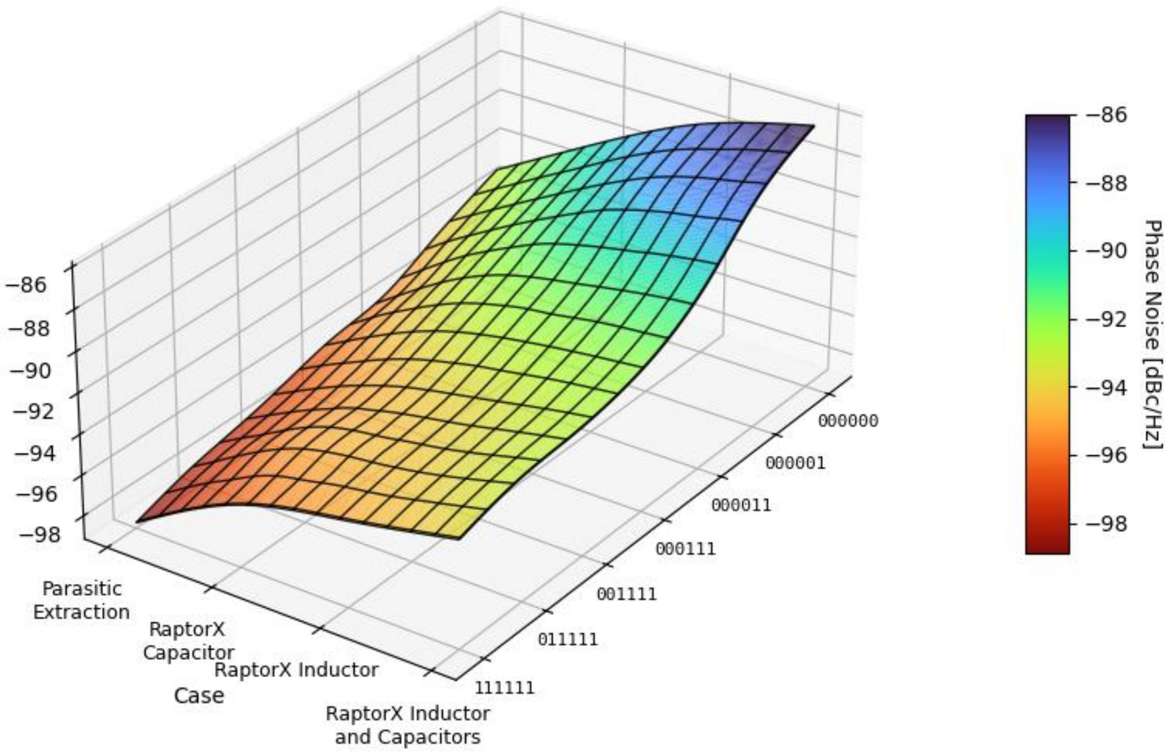


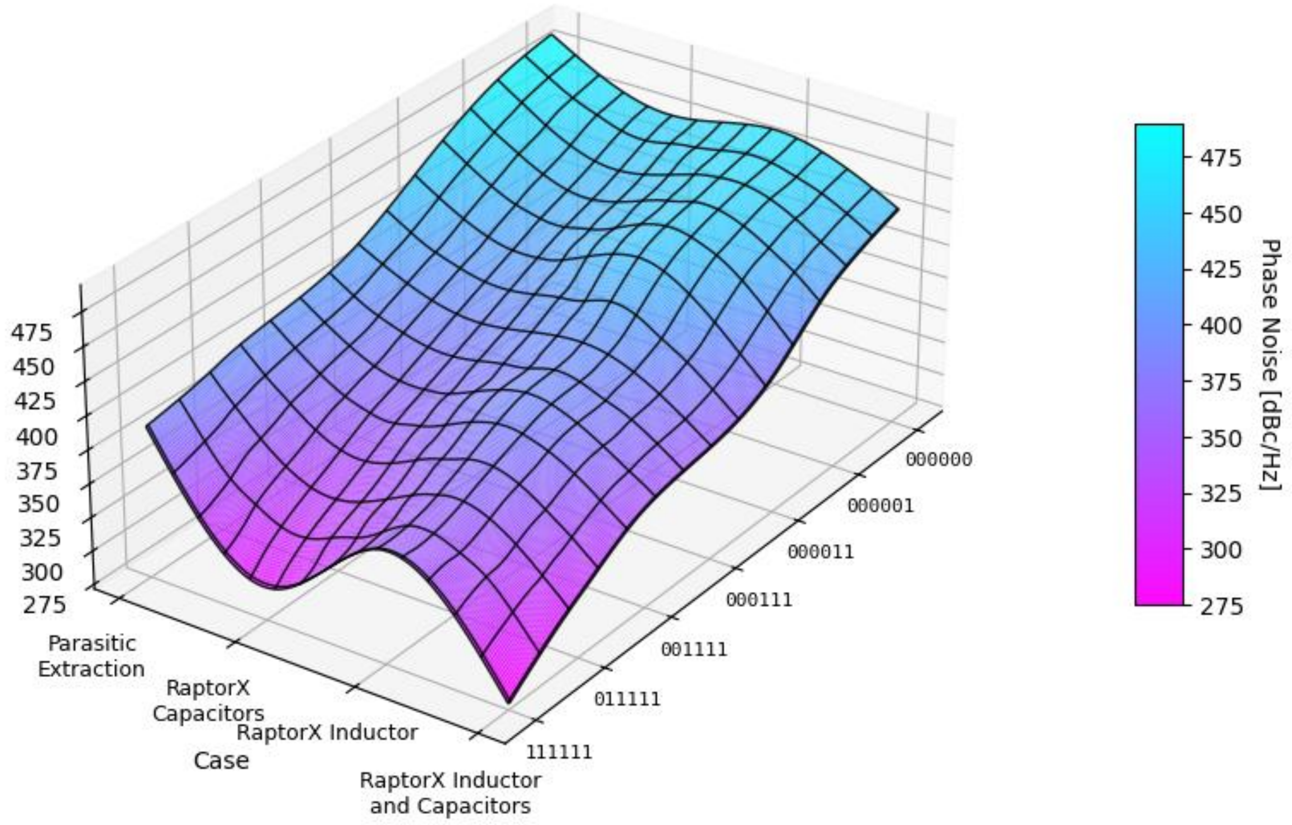
Layout Optimization



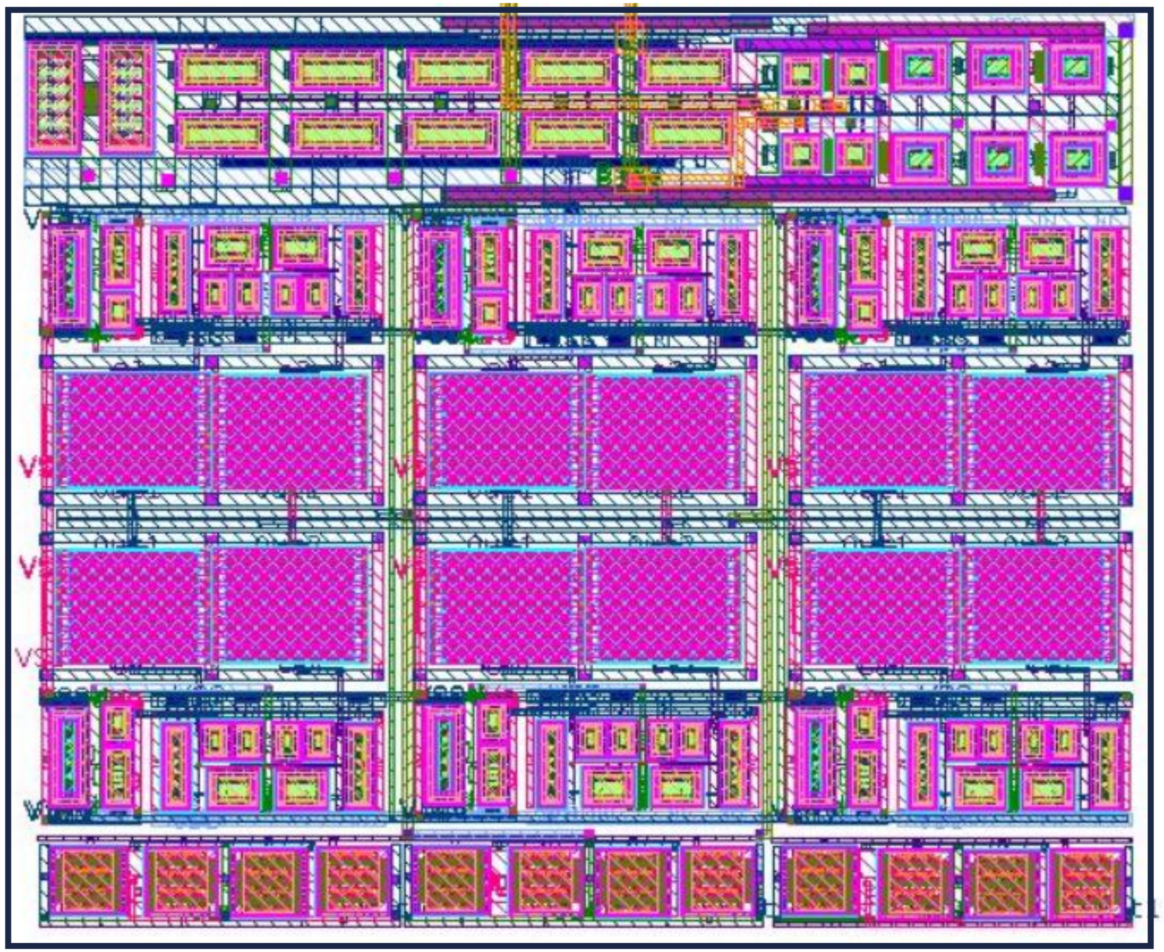


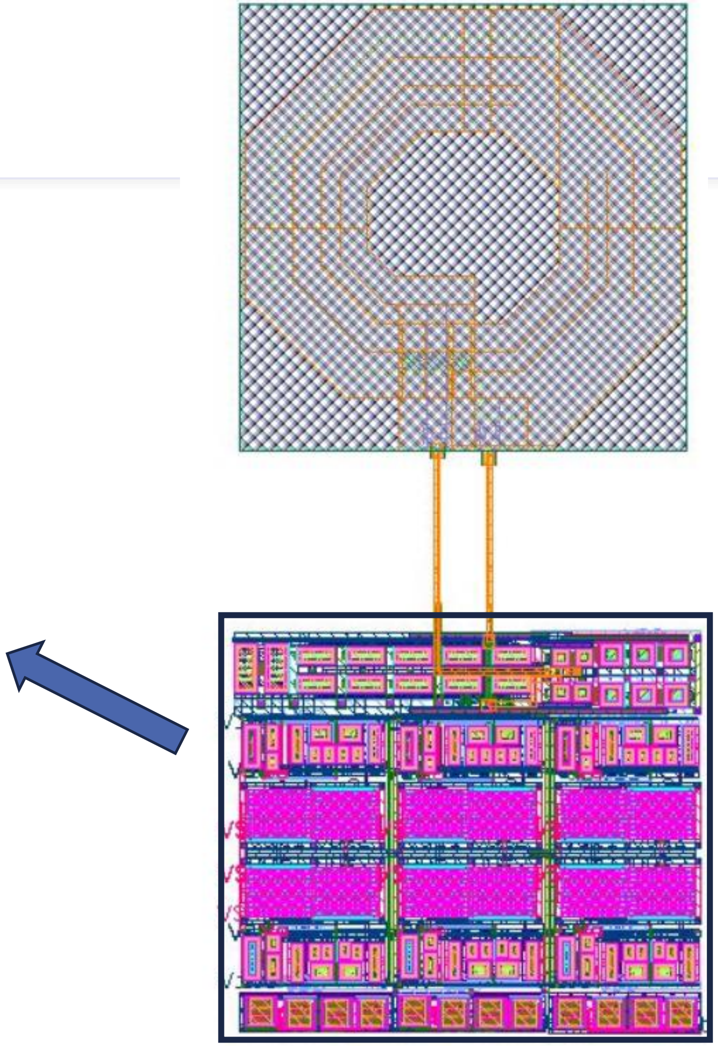
Post-Layout Results After Layout Optimization



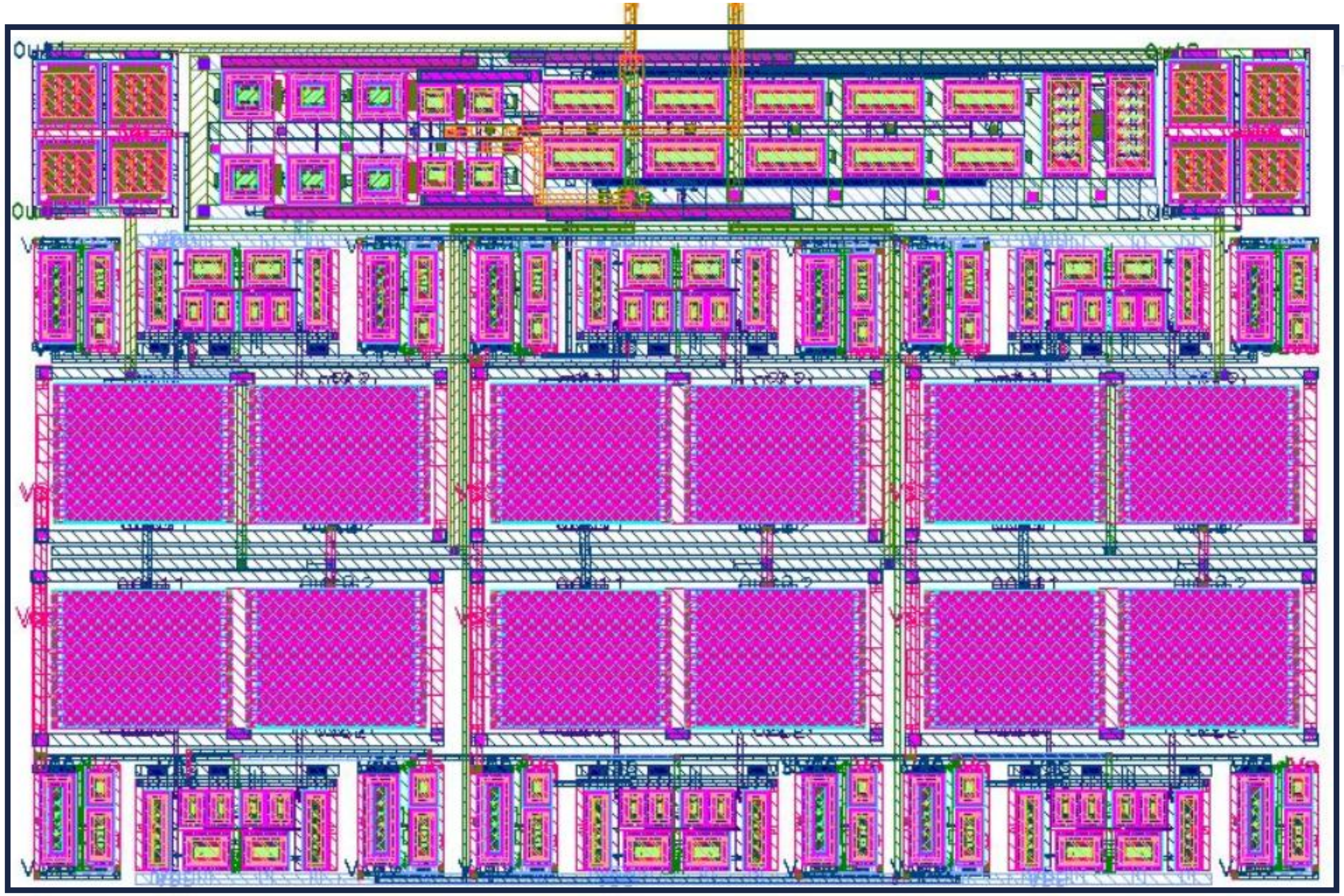


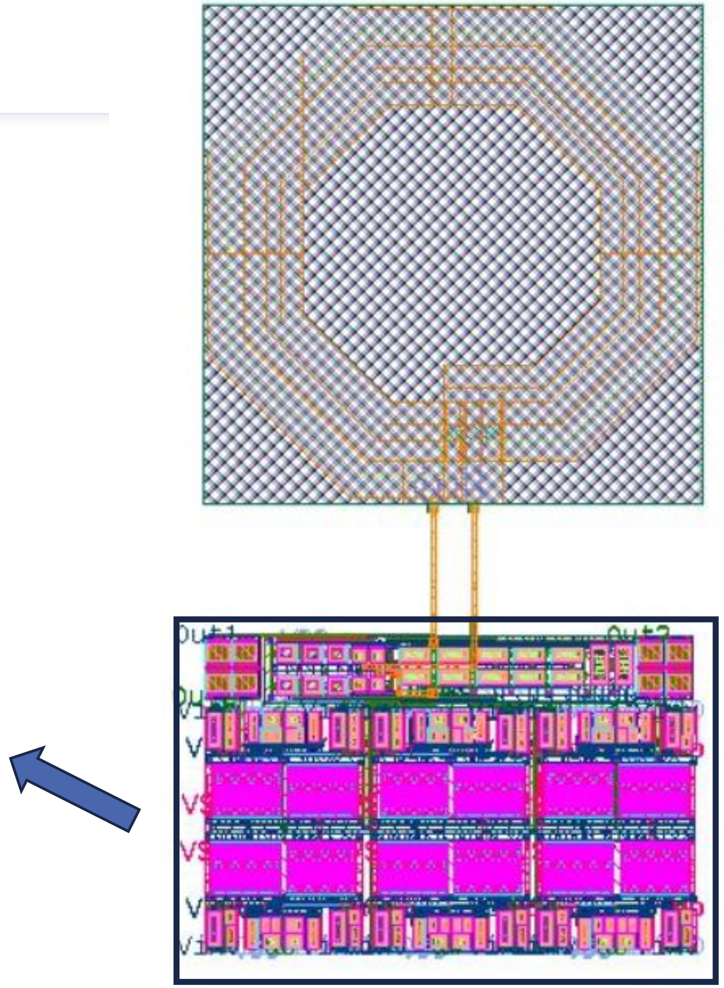
Middle-Frequency Band DCO Physical Design





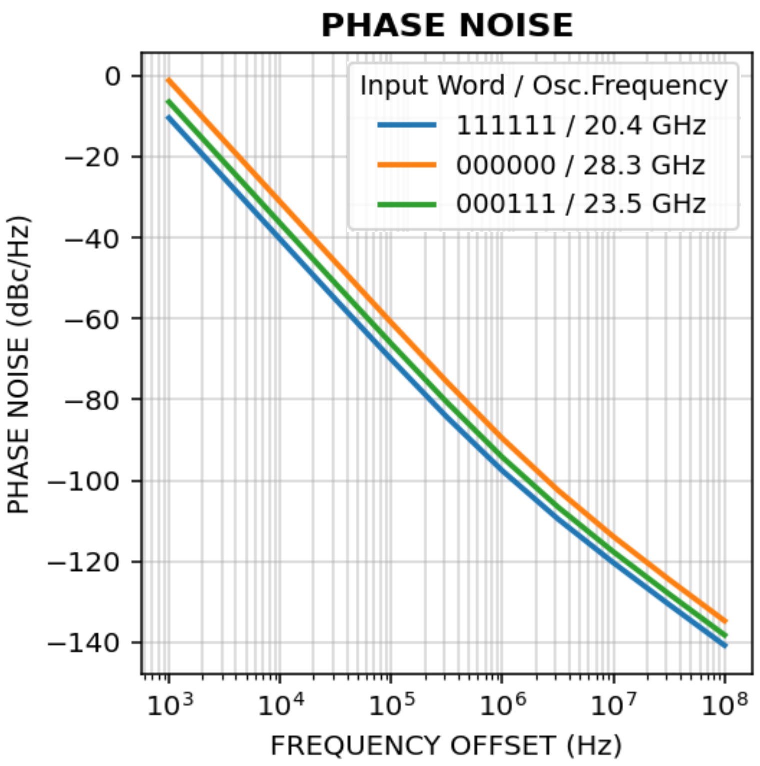
Low-Frequency Band DCO Physical Design



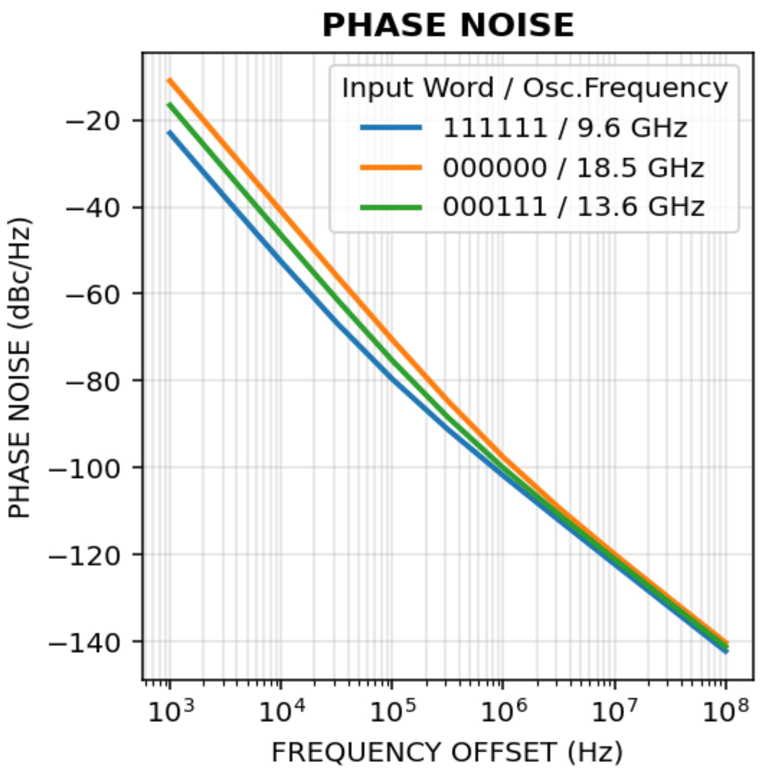


Post-Layout Simulation Results – Phase Noise

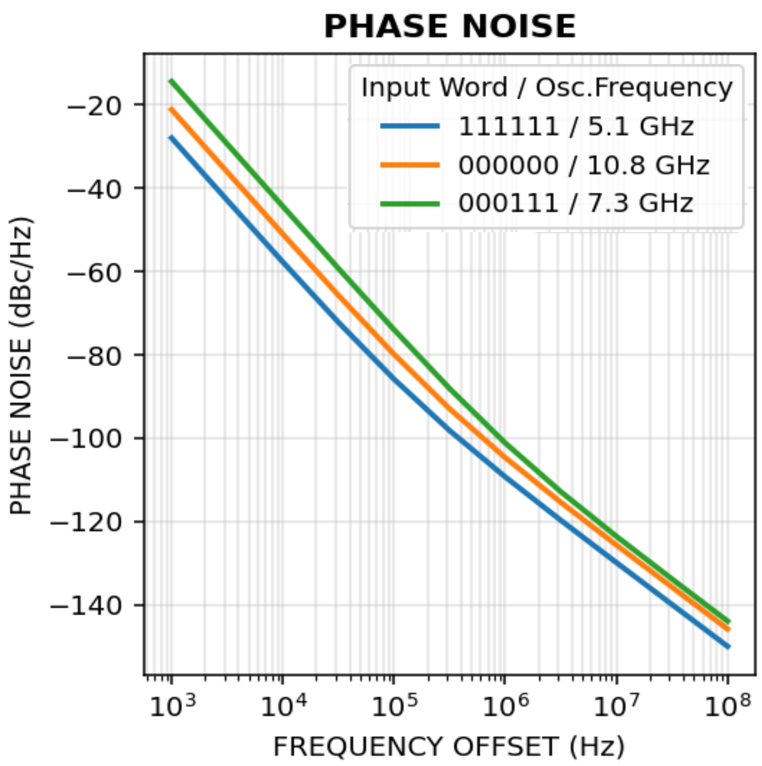
High Frequency Band



Middle Frequency Band



Low Frequency Band

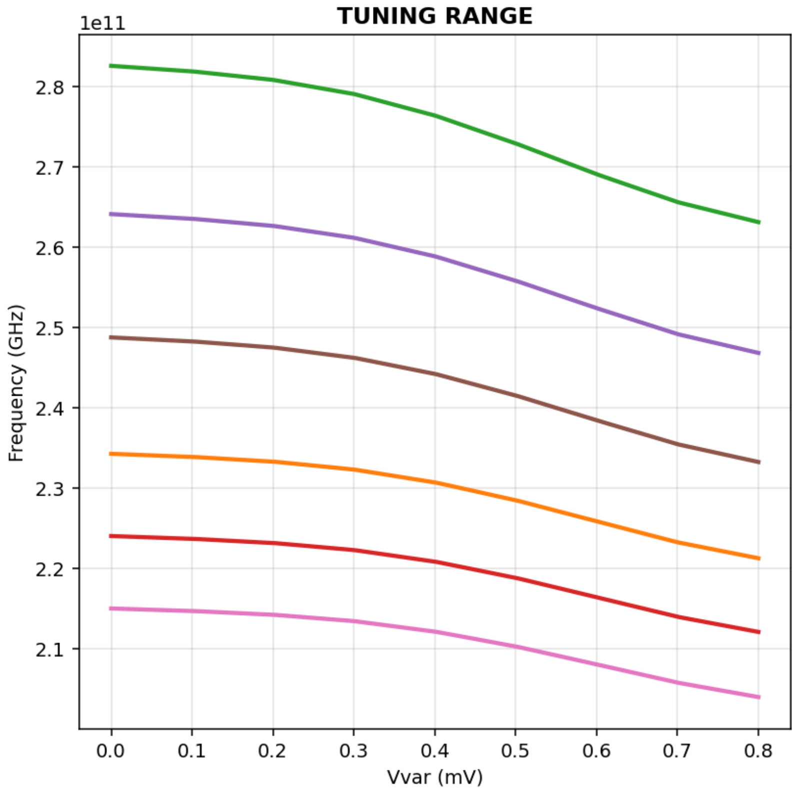


Phase Noise [dBc /Hz]

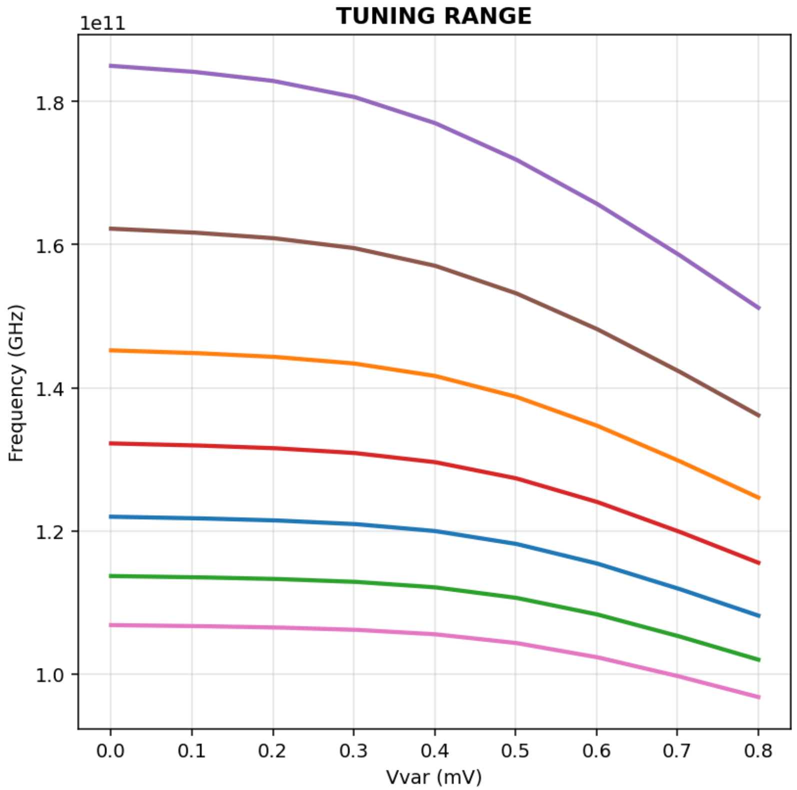
Input Word	HFB [20.2-28.3 GHz]	MFB [9.5-18.4 GHz]	LFB [5.1-10.8 GHz]
000000	-93	-97.6	-102.4
111111	-99.5	-101.8	-109.5

Post-Layout Simulation Results – Tuning Range

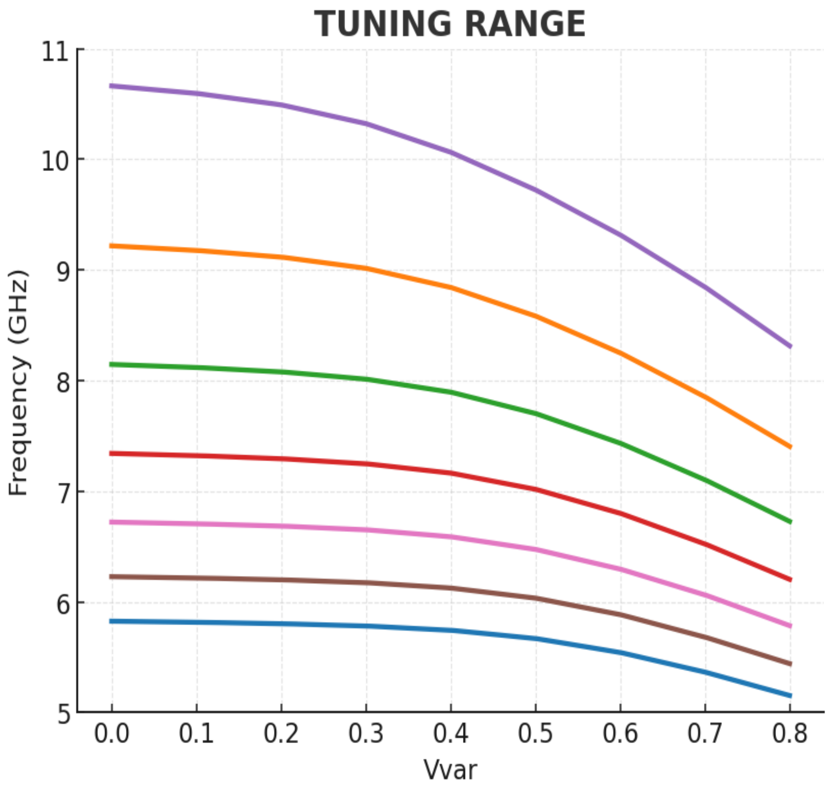
High Frequency Band



Middle Frequency Band



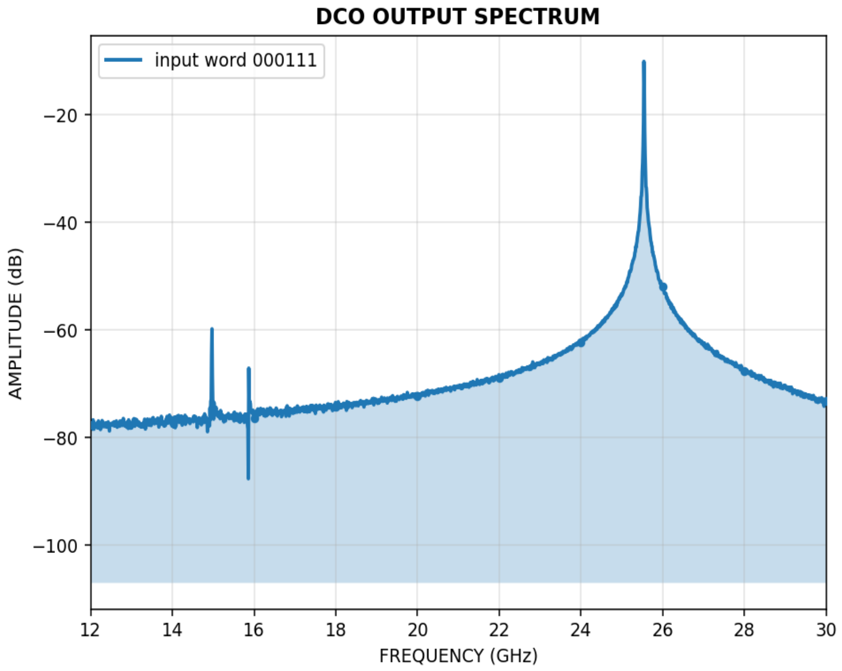
Low Frequency Band



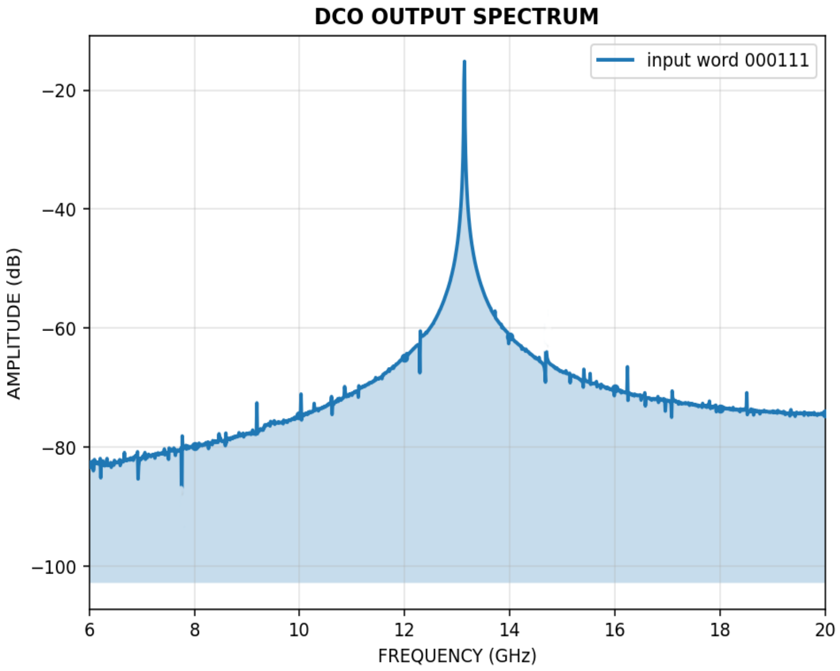
High Frequency Band	Middle Frequency Band	Low Frequency Band
8.1 GHz	8.9 GHz	5.7 GHz

Post-Layout Simulation Results - FFT

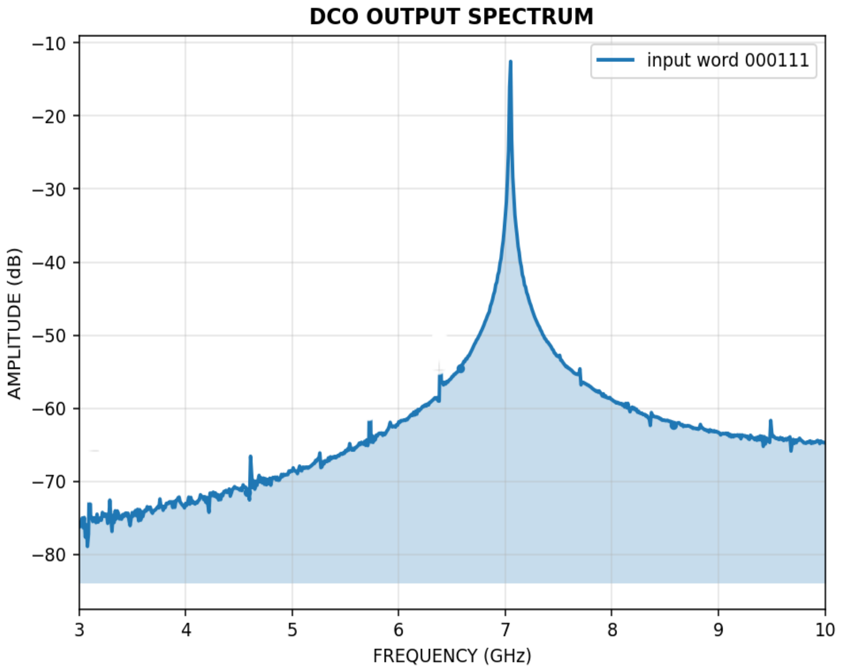
High Frequency Band



Middle Frequency Band



Low Frequency Band



Input Word	HFB Frequency [GHz]	MFB Frequency [GHz]	LFB Frequency [GHz]
000111	25.4	13.6	7.3

Comparison Table

Performance Metrics	HFB DCO	MFB DCO	LFB DCO	[1]	[2]	[3]	[4]	[5]	[6]
CMOS Process	22 nm FDSOI	22 nm FDSOI	22 nm FDSOI	65 nm	55 nm	22 nm FDSOI	22 nm FDSOI	28 nm	22 nm FDSOI
Center Frequency [GHz]	24.5	14.1	8	24	10	23.8	12	16.25	19.8
Tuning Range (%)	33.1	63.1	71.2	29	9	5	72	20.3	23.5
Phase Noise * [dBc/Hz]	-99.5	-101.8	-109.5	-104	-106.3	-96.6	-112.3	-117.3	-112.1
Power (mW)	8.5	7.8	7.3	16.9	7.3	14.4	33	22.1	24
Area (mm^2)	0.02	0.022	0.025	0.6	N/A	0.026	0.39	0.16	0.596
FoM_T [dBc/Hz]**	-188.4	-191.9	-196	-188.6	-176.8	-166.5	-195.8	-195.7	-192.1

*The phase noise was evaluated at a frequency offset of 1 MHz from the minimum oscillation frequency of the DCOs.

$$** \; FoM_T = PN - 20 \cdot log10(f_0 / \Delta f) + 10 \cdot log10(Pdiss / 1\, mW) - 20 \cdot log10(FTR / 10)$$

Conclusions

- ✓ The evaluation results demonstrate that the proposed *Machine Learning Framework for DCO Design and Phase Noise Optimization* is highly successful, achieving the generation of high-performance DCOs.
- ✓ The frequency-band division approach significantly improved the accuracy of both the frequency and phase noise prediction models, thereby contributing to a more precise and efficient DCO design process.
- ✓ It should be noted that the employed models were originally developed for VCO architectures, and their performance on DCOs is slightly reduced due to the omission of the LC tank's quality factor Q in the modeling process.

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